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Chapter 3

GELFAND-P000208

[source number: 1]

Year 1967. Gelfand Seminar.

18/IX. Kirillov.

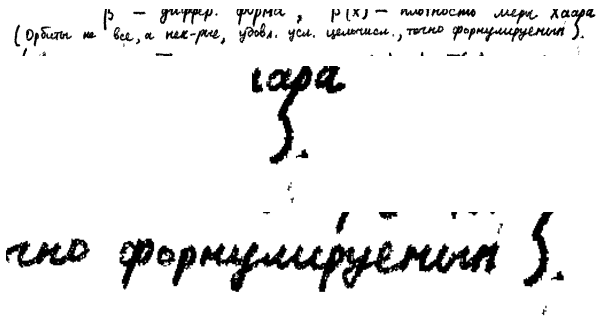
Formula (without proof or contradictory examples) for characters of irreduc. repres. of Lie groups.

$$\chi(x) = p(x)^{-1/2} \int_{\Omega} e^{i\langle F,x\rangle}\beta \qquad (\beta = \beta(F))$$

$x \in \mathfrak{g}$, $F \in \mathfrak{g}^*$, Ω — an orbit in $\mathfrak{g}^*$ under G,

β — a differ. form, $p(x)$ — the density of Haar measure.

(Not all orbits, but certain ones satisfying integrality cond., precisely [formulated: ending -емый / -емым?]}.



Uncertain reading. The final word ending is merged: “формулируемый” / “формулируемым” (formulated/formulable, with different endings). The closing sign has bends like a curly delimiter; candidate in the line: }.

Original source details: GELFAND-P000208

(G — a group, T — a unitary repres., $\chi(g) = \operatorname{tr} T(g)$ (in finite dim.)

In the general case χ — a gen. function

$$\langle \chi, \varphi\rangle = \int_G \varphi(g)T(g)\,dg$$ — already has a trace.

(Such a def. of the char.).)

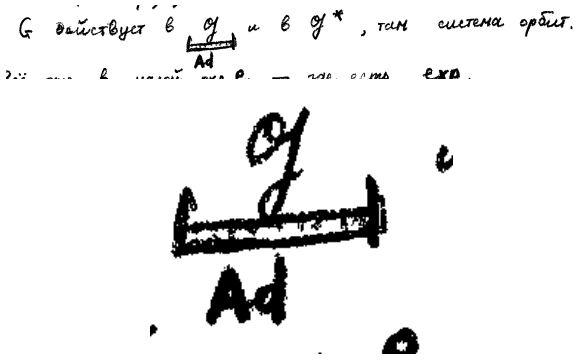


Diagram and typed description.

G acts in $\underset{\text{Ad}}{\mathfrak{g}}$ and in $\mathfrak{g}^*$; there is a system of orbits there.

Original source details: GELFAND-P000208

All this is in a small nbhd of e — where there is exp.

β is constructed from the struct. const. of the algebra.

[Crossed out: illegible.]



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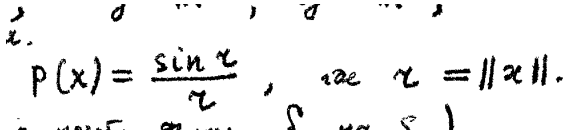
Undeciphered passage. One word beneath dense repeated loops. Undetermined candidates: “Предложения” (Proposals) / “Продолжение” (Continuation). The letters beneath the strokes are illegible.

Planned.

1. Expository talks on logic. (including the solution of the continuum hypothesis)
2. Lidskii “Asymptotics of eigenv. of self-adj. oper.”
3. Work of Kirillov; Kazhdan; Graev.

Example. $G = SO(3)$, $\mathfrak{g} = \mathbb{R}^3$, $\mathfrak{g}^* = \mathbb{R}^3$,
the action — rotations.
Orbit — S^2.

$$p(x) = \frac{\sin \tau}{\tau}, \qquad \text{where } \tau = \|x\|.$$

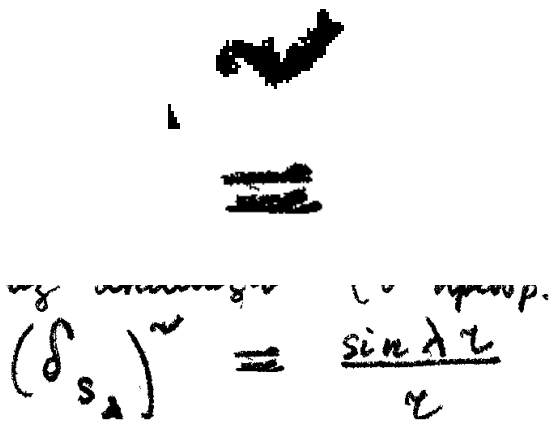


Original source details: GELFAND-P000208

Uncertain reading. The repeated radial-variable glyph has an upper stroke and a hooked leg: τ / r. Candidate in the line: τ; the literal letter is undetermined. No square on the ratio is distinguishable.

A fact from analysis (on the Fourier transform of δ on S_λ)

$$(\delta_{S_\lambda})^\vee \equiv \frac{\sin \lambda\tau}{\tau}$$



Original source details: GELFAND-P000208

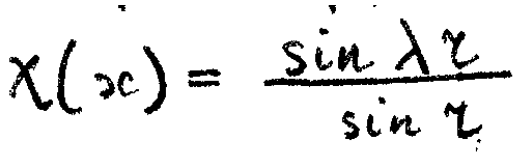
Uncertain reading. Above the closing parenthesis is a short two-stroke mark: $\sim$ / $\vee$. The densely inked sign between parenthesis and ratio: $\equiv$ / $=$. Candidates in the line: $\vee$, $\equiv$; neither reading is established. The radial variable is the same as in the preceding ratio: τ / r.

GELFAND-P000209

β — ordinary surface of the sphere.

Answer.

$$\chi(x) = \frac{\sin \lambda\tau}{\sin \tau}.$$



Original source details: GELFAND-P000209

Uncertain reading. The radial grapheme is the same as on the preceding page: τ / r. Candidate in the ratio: τ; the literal letter is undetermined.

All this for $\lambda = 2n + 1$ (countably many repres.)

$\beta(x)$ — $\int$ over cycles of the orbit of certain integral forms.

Work of Fuchs and Gelfand.

Fuchs. Study of principal bundles

[Crossed out: illegible.]



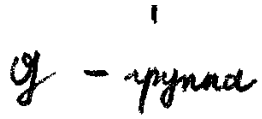
Original source details: GELFAND-P000209

Undeciphered passage. A short, densely crossed-out entry between “bundles” and “with fiber”. The letters are undetermined.

with fiber G (top. group)

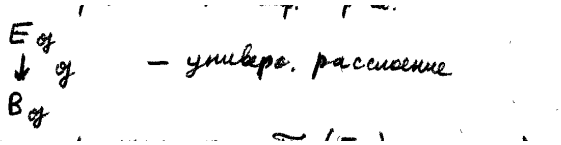
Types of invar. — as top. spaces and classif. spaces.

$\mathfrak{g}$ — a group



Original source details: GELFAND-P000209

Uncertain reading. The looped group sign: lowercase fraktur-like $\mathfrak{g}$ / handwritten uppercase G. Candidate $\mathfrak{g}$ is repeated in subscripts, arrow labels, and under the hat; the upright G above has a different form. The letter case is undetermined.



Original source details: GELFAND-P000209

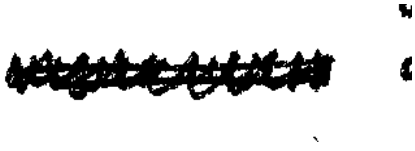
Diagram and typed description.

$E_{\mathfrak{g}}$

$\downarrow \mathfrak{g}$ — universal bundle

$B_{\mathfrak{g}}$

($E_{\mathfrak{g}}$ [Crossed out: illegible.]



Original source details: GELFAND-P000209

Undeciphered passage. One word beneath repeated dense strokes before “aspherical”. The letters are undetermined.

aspherical, i.e. $\pi_k(E_{\mathfrak{g}}) = 0$, $k > 0$).

A char. class is determ. by specifying $\alpha \in H^q(B_{\mathfrak{g}}; A)$

(Maps of principal bundles [перестан. / перестав. (commute?)] with the group action).



Original source details: GELFAND-P000209

Uncertain reading. The penultimate strokes before the period in the abbreviation “перестан.” / “перестав.” (commute?) are merged. Neither reading is established.

Analogously, instead of H^q — any contravar. functor h to the category of sets.

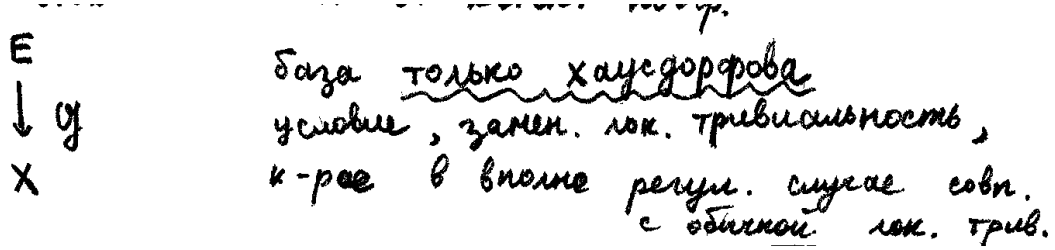
All char. classes — $h(B_{\mathfrak{g}})$

(this is if h — a homotopy functor)

$\hat{\mathfrak{g}}$ — a max. comp. subgroup

(all of them are homot. equiv. and their classif. spaces hom. equivalent.)
and equiv. to $\mathfrak{g}$.

G. Problem: find the actual classif. space of noncomp. groups, without identifying it with the max. comp. subgr.



Original source details: GELFAND-P000209

Typeset diagram

E (↓ $\mathfrak{g}$) X

base only Hausdorff
a condition replacing loc. triviality,
which in the completely regul. case coincides
with usual loc. triv.

GELFAND-P000210

[source number: 2]

2

The following property is required to hold:
consider the cone over $\mathfrak{g}$



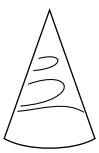
Original source details: GELFAND-P000210

Uncertain reading. The looped group sign is the same as on the preceding page: fraktur-like lowercase $\mathfrak{g}$ / handwritten uppercase G. Candidate in the lines and diagrams: $\mathfrak{g}$; the letter case is undetermined. The heading has a wavy underline.



Original source details: GELFAND-P000210

Uncertain reading. The outer cone outline, rounded base, and internal curved paths. Closure of the paths and possible arrow directions are undetermined.



$\mathfrak{g}$ acts on it with one fixed point.

$\forall x \in X$, $\exists$ a map $E \longrightarrow C\mathfrak{g}$, commuting

with the group action; the fiber over [Crossed out: an illegible sign.]



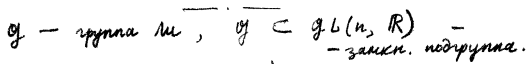
Original source details: GELFAND-P000210

Undeciphered passage. Before x a sign with a lower loop is densely crossed out. Undetermined candidates: y / g.

x maps isomorphically onto the base of the cone
(Local triviality follows from this).
(For a completely regul. base (there exists a function 1 at the point, 0 outside a given nbhd) loc. triviality $\Longrightarrow$ this property.)
(but in this case the theory becomes trivial and all computed invar. are trivial)

D. Kazhdan's remark: then the height of the cone determines a cont. function on X.
So this def. is incorrect.

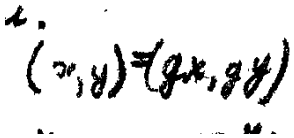
$\mathfrak{g}$ — a Lie group, $\mathfrak{g} \subset gL(n,\mathbb{R})$ — a closed subgr.



Original source details: GELFAND-P000210

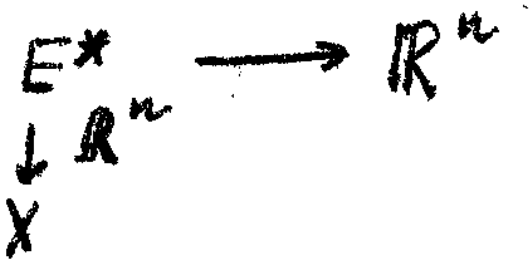
Uncertain reading. The first letter before L has a lower loop: $gL(n,\mathbb{R})$ / $GL(n,\mathbb{R})$. Candidate in the line: gL; the letter case is undetermined. The group sign itself is the same as in the cone heading.

(i.e. $\mathfrak{g}$ acts on $\mathbb{R}^n$).
There exists a construction which makes a vector bundle (induced) from a principal bundle
$E^* = E \times \mathbb{R}^n/\mathfrak{g}$.
$(x,y) = (gx, gy)$



Original source details: GELFAND-P000210

Uncertain reading. Between the pairs are two short dense strokes merged with the opening parenthesis of the right-hand pair: $=$ / $\sim$. Candidate in the line: $=$; the sign is undetermined.



Original source details: GELFAND-P000210

Diagram and typed description.

$$\begin{array}{ccc} E^* & \longrightarrow & \mathbb{R}^n \\ \downarrow{\scriptstyle \mathbb{R}^n} & & \\ X & & \end{array}$$

There must exist a map $E^* \longrightarrow \mathbb{R}^n$,

[Crossed out: illegible.]



Original source details: GELFAND-P000210

Undeciphered passage. A short word before "linear" is crossed out by two strokes. The letters are undetermined.

linear on the fibers and
such that at a certain point x_0 (given)
it — an isomorphism.
(No compatibility with the group action).



this is the def. of loc. triviality (for Hausd.) and this is precisely the requirement.
Kazhdan's remark cannot be repeated.

GELFAND-P000211

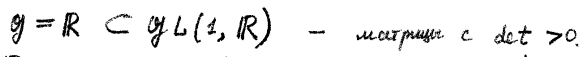
E is recovered from E^*.
One cannot construct a classif. space, but there is an analogue.
Example:
[An undetermined thickened sign.]



Original source details: GELFAND-P000211

Uncertain reading. Before the group notation is a separate thickened sign with a lower loop and a transverse stroke: "в" (in) / a crossed-out earlier sign. Neither candidate is established.

$\mathfrak{g} = \mathbb{R} \subset gL(1, \mathbb{R})$ — matrices with det > 0.



Original source details: GELFAND-P000211

Uncertain reading. The first letter before L has a lower loop: gL / GL. Candidate in the line: gL; the letter case is undetermined. The looped group sign itself is the same as on the preceding pages: $\mathfrak{g}$ / G, candidate $\mathfrak{g}$.

A nontrivial (principal) $\mathbb{R}$-bundle (also "universal").
The direct product of countably many lines in the Tych. topology.

$$(x_1, x_2, \ldots) \quad \text{— sequences}$$

$$\alpha(x_1, \ldots) = (x_1 e^{\alpha}, \ldots)$$

The quotient space S cannot be embedded in E (!) — — there are no sections.
Fact: in $\mathbb{R}^\infty$ one cannot choose one point on each ray.

Proof.

S is Hausdorff, but every cont. function on it is const,

so such a map — to a point (coords. — cont. functions).

(Show that with [рассет. / рассеч.? (section?)]



Original source details: GELFAND-P000211

Uncertain reading. After "with" the source abbreviation is "рассет." / "рассеч." (section?). The last letter has an upper stroke and a downward stroke; the reading is undetermined.

$$\sum \frac{1}{2^n} \frac{|x_i|}{1+|x_i|} = 1/2$$

one obtains a discontinuous map.)

Suppose there is f. (a cont. function).

$$(x_1, x_2, \ldots)$$

$$a_i = f(\underbrace{0, \ldots, 0}_{i}, 1, \ldots, 1, \ldots)$$

$$a = f(x_1, x_2, \ldots)$$

We shall prove that $a = \lim_{i\to\infty} a_i$.

$\forall \varepsilon > 0 \; \exists$ a nbhd of the point

[Crossed out: an illegible sign.]



Original source details: GELFAND-P000211

Undeciphered passage. Before the coordinate parenthesis a sign is densely crossed out. The original grapheme is undetermined.

$(x_1, x_2, \ldots) \in S$,

where the difference is $< \varepsilon$. (from a).

$$(x_1^0, x_2^0, \ldots)$$

(A neighborhood — if the preimage is open)

GELFAND-P000212

[source number: 3]

3

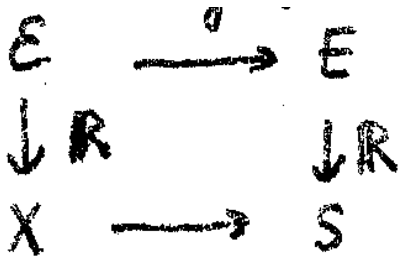
A neighborhood in E

$$|x_1 - x_1^0| < \delta$$
$$\ldots$$
$$|x_N - x_N^0| < \delta.$$

Make these coords. the same, the others very large.

$$\left(\frac{x_1^0}{M}, \ldots, \frac{x_N^0}{M}, 1, 1, \ldots\right),$$

and by these the point $(0, \ldots, 0, 1, \ldots, 1, \ldots)$ is approxim.
A map always exists (but is not unique)



Original source details: GELFAND-P000212

Uncertain reading. The upper-left vertex is rounded: calligraphic $\mathcal{E}$ / ε. Candidate in the diagram: $\mathcal{E}$; the letter is undetermined. Both horizontal arrows point right, both vertical arrows down with labels $\mathbb{R}$.

$$\begin{array}{ccc} \mathcal{E} & \longrightarrow & E \\ \downarrow{\scriptstyle \mathbb{R}} & & \downarrow{\scriptstyle \mathbb{R}} \\ X & \longrightarrow & S \end{array}$$

S is constructed from a product of $\mathbb{R}$ of any cardinality (depends on X)
$25/\overline{\text{IX}}$.
A good formula (Feynman)

$$\frac{1}{ab} = \int_0^1 \frac{dt}{(at + b(1-t))^2}$$

(decomp. into partial fractions)

$$\frac{1}{abc} = \int_\Delta \frac{dt_1\, dt_2}{(at_1 + bt_2 + ct_3)^3},$$

Δ — a simplex

$$t_1 + t_2 + t_3 = 1, \quad t_i \geq 0.$$

Next time Fuchs and then logic.
There is a counterexample to the hypothesis of a common fixed point of commuting cont. maps of an interval, and it is proved that if they are differentiable, there is a point fixed for one and of finite multiplicity for the other.

GELFAND-P000213

Fuchs (continuation).
$\mathfrak{g} \subset gL(n; \mathbb{R})$



Original source details: GELFAND-P000213

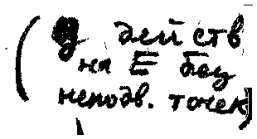
Uncertain reading. The first letter before L has a lower loop: gL / GL; a semicolon follows n. Candidate in the line: gL. The group sign itself is the same as before: $\mathfrak{g}$ / G, candidate $\mathfrak{g}$; the letter case is undetermined.

principal bundles

$$\begin{array}{ll} (\xi) & p : E \longrightarrow X \\ (\xi^\#) & p^\# : E^\# \longrightarrow X \end{array}$$

(X — Hausdorff)

($\mathfrak{g}$ acts on E without fixed points]



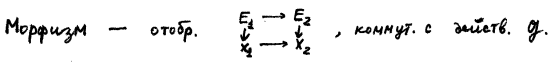
Original source details: GELFAND-P000213

Uncertain reading. The right-hand entry about an action without fixed points. The closing sign is angular:] / ⟩. Candidate in the line:]; the reading is undetermined.

(

$$xg_1 = xg_2 \Rightarrow$$
$$\Rightarrow g_1 = g_2$$

Loc. triviality: $\forall x \; \exists$ a map $E^{\#} \longrightarrow \mathbb{R}^n$, an isom. at the point x, linear on the fibers.
(One cannot require 0 outside small nbhds, since then even $X \times \mathbb{R}^n$ may fail to qualify because of nonregularity)



Original source details: GELFAND-P000213

Diagram and typed description.

A morphism — a map

$$\begin{array}{ccc} E_1 & \longrightarrow & E_2 \\ \downarrow & & \downarrow \\ X_1 & \longrightarrow & X_2 \end{array}$$

, commuting with the action of $\mathfrak{g}$.

Examples of functors:

1) cohomology

2) H-bundles (H — another group)

correspondingly charact. classes.

$h_{\text{alg}}(\mathfrak{g})$ — the set (group) of all char. classes constructed from the functor h.

1) For any top. space

[Crossed out: an illegible sign.]



Original source details: GELFAND-P000213

Undeciphered passage. After “пр-ва” (space), before X, an additional sign is densely crossed out. The original grapheme is undetermined.

X, the sheaf $\mathbb{R}$ of germs of cont. real functions on X is defined.

Put

$$h(X) = H^q(X, \mathbb{R}).$$

(for [Crossed out: illegible.]



Original source details: GELFAND-P000213

Undeciphered passage. A fragment between “for” and “paracomp.” is crossed out; a descender is visible. The letters are undetermined.

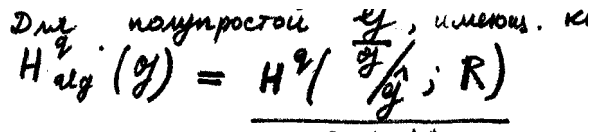
paracomp. X this is 0)

There is defined

$$H^q_{\text{alg}}(\mathfrak{g})$$

Answer. For semisimple $\mathfrak{g}$ having finitely many connected components

$$H^q_{\text{alg}}(\mathfrak{g}) = H^q(\bar{\mathfrak{g}}/\hat{\mathfrak{g}}; R)$$



Original source details: GELFAND-P000213

Uncertain reading. Above the first right-hand group sign is a straight bar, above the second an angular hat; a slash separates them. Candidate in the line: $\bar{\mathfrak{g}}/\hat{\mathfrak{g}}$. A descender from the preceding line overlaps the upper strokes; the reading of the first superscript mark remains open. The whole right-hand side is underlined.

with coeffs. in a constant sheaf —
— real coeffs.

GELFAND-P000214

[source number: 4]

4

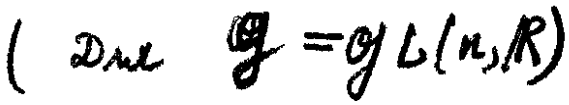
Here $\bar{\mathfrak{g}}$ — a compact form of $\mathfrak{g}$
$\hat{\mathfrak{g}}$ — a max. comp. subgroup
[Crossed out: $\bar{\mathfrak{g}}$?]



Original source details: GELFAND-P000214

Undeciphered passage. Before the final group sign an entry is densely crossed out. Candidate $\bar{\mathfrak{g}}$ from the visible upper bar and lower loop; the precise form beneath the strokes is undetermined.

$\mathfrak{g}$. (For $\mathfrak{g} = gL(n, R)$



Original source details: GELFAND-P000214

Uncertain reading. The initial letter before L matches the looped lowercase group sign: gL / GL. Candidate in the line: gL; the letter case is undetermined. The group sign itself is the same as before: $\mathfrak{g}$ / G, candidate $\mathfrak{g}$.

there will be $\bar{\mathfrak{g}} = U(n)$, $\hat{\mathfrak{g}} = O(n)$
$\bar{\mathfrak{g}}$ is obtained as a max. comp. subgroup of the complexification of $\mathfrak{g}$)
2nd order: as for S^2.
For a compact group everything is trivial.
For a nonsemisimple one, take $\mathfrak{g}/\hat{\mathfrak{g}}$ and differ. forms on $\mathfrak{g}/\hat{\mathfrak{g}}$ invariant under $\mathfrak{g}$.
d(invar.) — invar. And the homologies of this complex are isomorphic to $H^q_{\text{alg}}(\mathfrak{g})$.

$$H^q_{\text{alg}}(\mathbb{R}; \mathbb{R}) = \begin{cases} R & q = 0, 1 \\ 0 & q > 1. \end{cases}$$

Check! (there are 1 and dx)

2) X, H — a top. group

$EH(X)$ — the set of all loc. triv. (in the usual sense) bundles (principal) on X.

\

Theorem.

$$EH_{\text{alg}}(\mathfrak{g}) = \text{Hom}(\mathfrak{g}, H)$$

(cont. hom. [Crossed out: an illegible sign.]



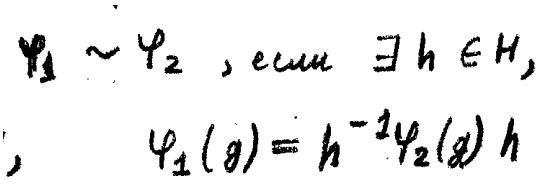
Original source details: GELFAND-P000214

Undeciphered passage. After “cont. hom.” are doubled curved and oblique strokes: a cancelled) / the sign φ. The reading is undetermined.

, more precisely, classes of conjugate homom.)

The natural answer — a hom. $\mathfrak{g} \longrightarrow H$ corresponds to a char. class, naturally defined.

$$\varphi_1 \sim \varphi_2, \text{ if } \exists h \in H, \qquad \varphi_1(g) = h^{-1}\varphi_2(g)h$$

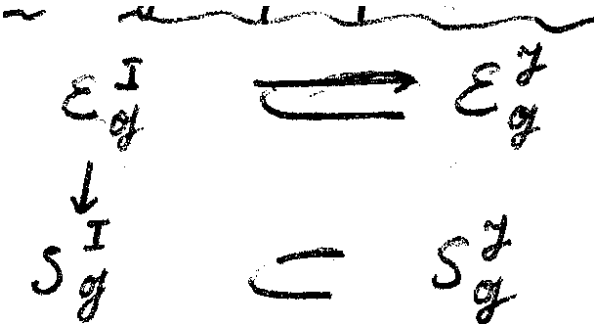


Original source details: GELFAND-P000214

Uncertain reading. The map signs have diverging upper strokes and a long stem: φ / ψ. Candidate in the line: φ, in both indexed names. The exact Greek letter is undetermined. The sign $\sim$, h^{-1}, and order of factors are readable.

GELFAND-P000215

Construction of a universal bundle.



Original source details: GELFAND-P000215

Uncertain reading. At the top is an elongated inclusion with a rightward arrowhead on the upper stroke. The precise reading as $\hookrightarrow$ / $\subset$ with an added arrow is undetermined. There is a downward arrow on the left and $\subset$ below; there is no right vertical arrow. The rounded upper letter is rendered by candidate $\mathcal{E}$.

$$\begin{array}{ccc} \mathcal{E}^I_{\mathfrak{g}} & \xrightarrow{\subset} & \mathcal{E}^J_{\mathfrak{g}} \\ \downarrow & & \\ S^I_{\mathfrak{g}} & \subset & S^J_{\mathfrak{g}} \end{array}$$

$I \subset J$ — arbitrary sets.

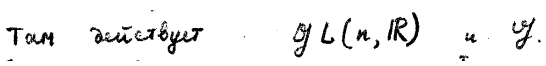
Any bundle maps into a universal one with I of sufficiently large cardinality.

Consider $\mathbb{R}^I \ni (x_i,\ i \in I)$.

The topology — Tychonoff.

and $\underset{n \text{ lin. indep. vect.}}{n\text{-frames}}$ in it — this is precisely $\mathcal{E}^I_{\mathfrak{g}}$.

$gL(n, \mathbb{R})$ and $\mathfrak{g}$ act there.



Original source details: GELFAND-P000215

Uncertain reading. The initial letter before L matches the lower looped group sign on the right: gL / GL. Candidate in the line: gL; the letter case is undetermined. The group sign itself is rendered by the same candidate $\mathfrak{g}$ as before.

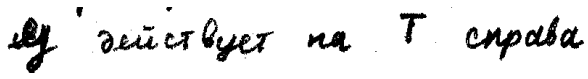
The quotient space is precisely our $S^I_{\mathfrak{g}}$.
Observe that $\mathcal{E}^I_{\mathfrak{g}}$ does not depend on $\mathfrak{g}$.
Lemma. $\forall \mathcal{E}^I_{\mathfrak{g}}$ and a cont. function f on it $\exists$
$J \subset I$ such that $I \setminus J$ is at most countable, and
and $f|_{\mathcal{E}^J_{\mathfrak{g}}} =$ const.

GELFAND-P000216

[source number: 6]

6

$2/\overline{\text{X}}$. Fuchs.
T — a space, $S \subset \text{Map}(T, T)$, compact open top.
$\mathfrak{g}$ acts on T on the right

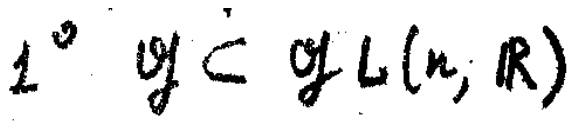


Original source details: GELFAND-P000216

Uncertain reading. The group sign is looped: lowercase fraktur-like $\mathfrak{g}$ / handwritten uppercase G. Candidate in the lines, subscripts, and diagrams: $\mathfrak{g}$. The element sign g in the next line is simpler; the letter case of the group sign is undetermined.

$$f \in S, \quad g \in \mathfrak{g} \Rightarrow fg \in S,\ gf \in S.$$

Examples.
1°. $\mathfrak{g} \subset gL(n; R)$,



Original source details: GELFAND-P000216

Uncertain reading. In the first example $gL(n; R)$ / $GL(n; R)$; the initial letter is looped. Candidate in the line: gL; the letter case is undetermined.

$T = \mathbb{R}^n$; S — all lin. maps (including zero maps).
2°. $\mathfrak{g}$ — a Hausdorff top. group, $T = C\mathfrak{g}$ — the cone over $\mathfrak{g}$



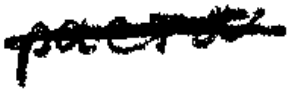
Original source details: GELFAND-P000216

Diagram and typed description.



$\mathfrak{g}$ acts naturally, S is generated by them and similarities

[Crossed out: растя... / растяже...? (dila...?)]



Original source details: GELFAND-P000216

Undeciphered passage. Before “contractions” a word or beginning of a word is crossed out; the opening “раст...” is distinguishable. The ending is undetermined.

contractions toward the vertex (and to a point).

Axioms.

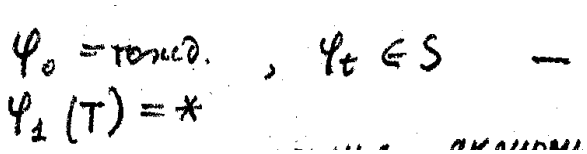
1°. T is contractible by transformations in S to a marked point $*$.

2°. $\mathfrak{g}$ is closed in the group of all homeom. of T to itself.

3°. $*$ is fixed under all transf. in S.

4°. $\varphi_t : T \longrightarrow T$

$$\varphi_0 = \text{identity}, \quad \varphi_t \in S \quad —$$
$$\varphi_1(T) = *$$



Original source details: GELFAND-P000216

Uncertain reading. The Greek map sign has diverging upper strokes and a stem: φ / ψ. Candidate in the fourth axiom and following phrase: φ; the exact letter is undetermined. Subscripts 0, t, 1, identity, and image $*$ are readable.

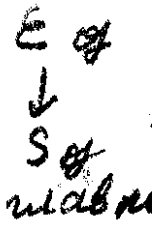
— the homotopy of axiom 1°.

Then there exist φ_t such that φ_t for $t < 1$ are topological embeddings.

5°. S contains all transf. of $\mathfrak{g}$, and maps close to maps of $\mathfrak{g}$ in S are top. embeddings.

$U(1) \subset S$ consists of embeddings.

In this situation spaces are defined



Original source details: GELFAND-P000216

Diagram and typed description.

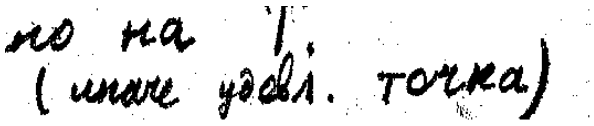
$$\begin{array}{c} \mathcal{E}_{\mathfrak{g}} \\ \downarrow \\ S_{\mathfrak{g}} \end{array}$$

, the action of $\mathfrak{g}$ on $\mathcal{E}_{\mathfrak{g}}$ — a universal principal bundle.

$$\underbrace{T \times \cdots \times T}_{I \text{ times}}$$ Tych. topology.

6°. $\mathfrak{g}$ acts effectively on T,

(otherwise [удвл. / удовл.?] point)

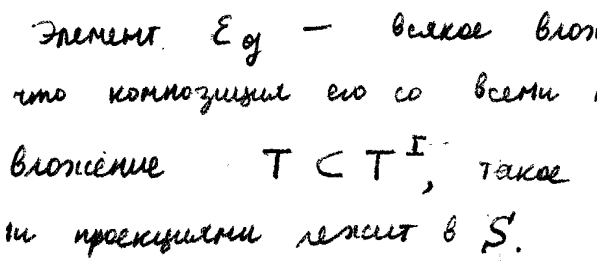


Original source details: GELFAND-P000216

Uncertain reading. The final parenthesis contains the abbreviation “удвл.” / “удовл.”. The reading and expansion of the word are undetermined.

GELFAND-P000217

An element of $\mathcal{E}_{\mathfrak{g}}$ — any embedding $T \subset T^I$ such that its compositions with all projections lies in S.



Uncertain reading. In the subscript of the rounded E and subsequently in $C\mathfrak{g}$, the action, and cohomology, the group sign is looped: $\mathfrak{g}$ / handwritten uppercase G. Candidate: $\mathfrak{g}$; the letter case is undetermined.

Original source details: GELFAND-P000217

1) $T = \mathbb{R}^n$ — all n-frames.

2) $T = C\mathfrak{g}$, $\mathcal{E}_{\mathfrak{g}} = C\mathfrak{g} \times \ldots \times C\mathfrak{g} \setminus \{*, \ldots, *\}$.

$\mathfrak{g}$ acts on $\mathcal{E}_{\mathfrak{g}}$ by multiplication by transf. in $\mathfrak{g}$.

I finite $\Rightarrow$ 1) the usual classif. space
2) Milnor's constr. with a not very substantial modification.



Diagram and typed description.

$\mathfrak{g} = \mathbb{R}$, $T =$ •———

Original source details: GELFAND-P000217

$\mathcal{E}_{\mathfrak{g}}$ — sequences of nonneg. numbers.

For each T its own definition of loc. triviality.

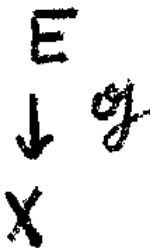


Diagram and typed description.

$$\begin{array}{c} E \\ \downarrow \mathfrak{g} \\ X \end{array}$$

Original source details: GELFAND-P000217

is loc. triv. if the induced bundle



Diagram and typed description.

$$\begin{array}{c} E^{\#} \\ \downarrow T \\ X \end{array}$$

Original source details: GELFAND-P000217

, has the property: $\forall x \; \exists$ a map $E^{\#} \longrightarrow T$, a homeom. on the fiber over x, and on $\forall$ fiber it is a transf. in S.

Theorem.

Every bundle maps into this one.

(X — all Hausdorff)

Defined $H^q_{\text{alg}}(\mathfrak{g};\mathbb{R})$

Theorem.

Whatever T and S may be, $H^q_{\text{alg}}(\mathfrak{g},\mathbb{R})$ does not depend on their choice

(Instead of $\mathbb{R}$ one may take a group A)

What will there be?

Uspenskii, Lectures on logic 2

Lidskii, Asymptotics of eigenfns. (after Hörmander)

Sinai, Conditions on diffeom. under which there is an invar. measure

Kazhdan, Discr. groups — solution of Selberg's problem

Kushnirenko, Analyt. action of a semisimple group in a nbhd. of a fixed point — linear

GELFAND-P000218

[source number: 7]

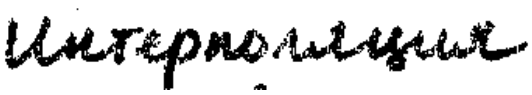
Tyurina and S. Gelfand survey of Pham's works.

Anosov and Maslov

I. Bernstein P^λ

Gelfand, Graev

S. G. Krein



Original source details: GELFAND-P000218

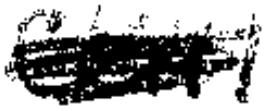
Uncertain reading. Интерполяции / Интерполяция (interpolation); the ending is uncertain.

Berezin
Kirillov' Grassmann analysis.

— • —

GELFAND-P000219

$16/\overline{\text{X}}$

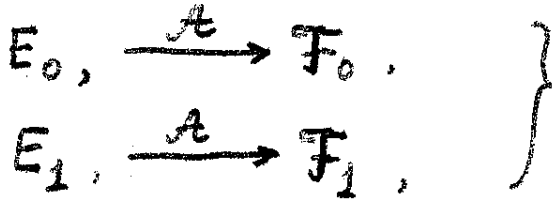


Original source details: GELFAND-P000219

Uncertain reading. [Crossed out: illegible.]

S. G. Krein.

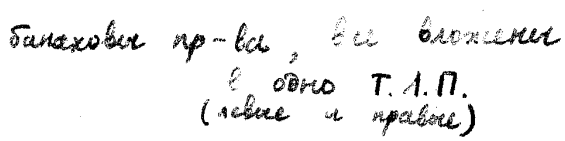
On interpolation of linear operators.



Original source details: GELFAND-P000219

Diagram and typed description.

$$\left.\begin{array}{l} E_0, \xrightarrow{A} F_0, \\ E_1, \xrightarrow{A} F_1, \end{array}\right\}$$ Banach spaces,



Uncertain reading. all embedded / all are embedded

Original source details: GELFAND-P000219

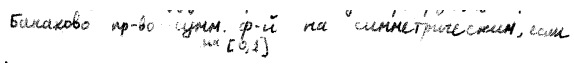
into one T. L. S. (the left and the right ones)

A is bounded.

E, F — an interpolation pair if, for $\forall$ A, boundedness at 0 and 1 implies boundedness of $A : E \longrightarrow F$.

All spaces will be spaces of summable functions.

A Banach space of summable functions [Inserted below the line: on $[0, 1]$.]



Uncertain reading. на / наз. (on / called)

Original source details: GELFAND-P000219

symmetric if

1) $y(t) \in E$

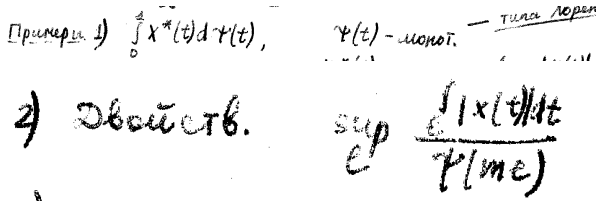
$$|x(t)| \le |y(t)| \quad \Longrightarrow \quad x \in E \qquad \text{and} \qquad \|x\| \le \|y\|$$

2) The norm does not change under transformations preserving measure (one may consider an equimeasurable monotone one)

Then $L_\infty \subset E \subset L_1$.

Examples. 1)

$$\int_0^1 x^*(t)\, d\psi(t), \qquad \psi(t) \text{ — monot.} \qquad \text{— of Lorentz type}$$



Uncertain reading. In the differential, the monotonicity label, and the denominator of the next example, the Greek sign has three upper arms and a lower stroke: ψ / Ψ. Candidate: ψ; the letter case is undetermined.

Original source details: GELFAND-P000219

$x^*(t)$ — a rearrangement of $|x(t)|$

2) Dual. $\sup\limits_e \dfrac{\int_e |x(t)|\, dt}{\psi(me)}$

3) and Orlicz spaces (these include L_p)

[At the left above the line:] 1st step.

Riesz–Thorin theorem.

GELFAND-P000220

2nd step.

Marcinkiewicz theorem.

(Less is required at the endpoints)

$$n_x(\tau) = \text{mes}\{t : |x(t)| \ge \tau\}$$

$$[n_{Ax}(\tau)]^{1/q_i} \le \frac{C}{\tau} \|x\|_{L_{p_i}}$$

— a less restrictive condition.

Most operators of analysis satisfy the Marcinkiewicz conditions at the endpoints

Matsaev and Dikarev — more precisely, from L_p into a space narrower than L_q (of Lorentz type)! — and it is already minimal (in the middle)

(Minimal for all operators under conditions both of Marcinkiewicz and of Riesz–Thorin) —

— this is for improving operators:

$$\begin{matrix} q_1 > p_1 \\ q_2 > p_2 \end{matrix} \quad \| \quad \text{— as in the Marcinkiewicz theorem.}$$

$p = q$ — maps into itself — then all the other scales do too.

Let $\begin{matrix} L_1 \longrightarrow L_1 \\ L_\infty \longrightarrow L_\infty \end{matrix}$ and $\|A\| \leq 1$

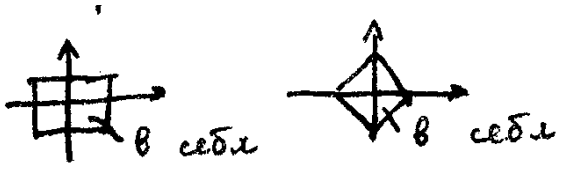
List all



Uncertain reading. [Crossed out: illegible.]

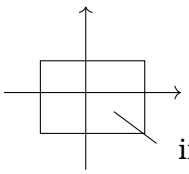
Original source details: GELFAND-P000220

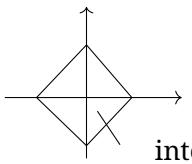
spaces for which $\|A\| \leq 1$ as well.



Original source details: GELFAND-P000220

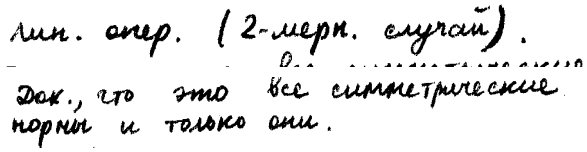
Diagram and typed description. The purpose and direction of the additional oblique strokes are undetermined.


into itself


into itself

[To the right of the figures:] linear oper. (2-dimensional case).

Док. [prove / proved], that these are all symmetric norms and only they.



Uncertain reading. The abbreviation “Док.” is readable; the full verb form is not written. Possible expansions: prove / proved.

Original source details: GELFAND-P000220

General theorem — B. S. Mityagin.

It is necessary and sufficient that the space be symmetric.

(also in the ∞-dimensional case)

$L_p \xrightarrow{A} L_p$ what then?

$p \neq 1, \infty$

(Marcinkiewicz case)

$$\| \underset{\substack{| \\ \text{char. fn. } e}}{\chi_e} \|_E = \varphi(me).$$

(symmetry is again necessary)

the resulting function $\varphi(t)$ — a characteristic of the space E.

$$\ell_E = \underline{\lim}_{t\to 0}\frac{\varphi(2t)}{\varphi(t)} \qquad u_E = \overline{\lim}_{t\to 0}\frac{\varphi(2t)}{\varphi(t)} \qquad 1 \le \ell_E \le u_E \le 2$$

GELFAND-P000221

[source number: 9]

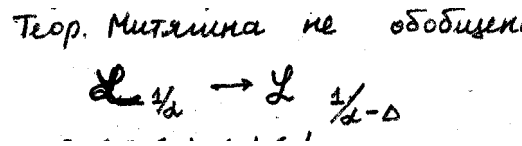
For L_p they are both equal to $2^{1/p}$

Boundedness of all such A in $E \Longleftrightarrow 1 < \ell_E \le u_E < 2$

(Semyonov's 1st theorem)

Mityagin's theorem has not been generalized.

$$L_{1/\alpha} \longrightarrow L_{1/\alpha-\Delta}$$



Original source details: GELFAND-P000221

Uncertain reading. The right subscript is written as a diagonal fraction without parentheses: $1/\alpha - \Delta$ / $1/(\alpha - \Delta)$. The grouping is undetermined.

$0 < \Delta \le \alpha_0 < \alpha < \alpha_1$.

How can one list the spaces in which operators of this type are bounded?

$$2^{\alpha_0} < \ell_E \le u_E < 2^{\alpha_1} \quad \Longrightarrow \quad E \longrightarrow E_\Delta,$$

where



Original source details: GELFAND-P000221

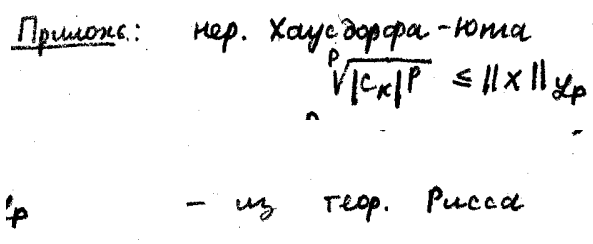
Uncertain reading. After "where" there are two dense short signs; their exact reading and whether they are crossed out are undetermined.

$$\|x\|_{E_\Delta} = \|x^*(t)t^{-\Delta}\|_E$$

(This generalizes the theorem of Dikarev and Matsaev — the best space)

Applic.: Hausdorff–Young inequality

$$\sqrt[p]{|c_k|^p} \le \|x\|_{L_p} \qquad \text{— from Riesz's theorem}$$



Original source details: GELFAND-P000221

Uncertain reading. The first radicand visibly contains $|c_k|^p$; no separate summation sign is distinguishable. The root index is p, and the right-hand norm is in L_p.

Poia inequality:

$$\sqrt[p]{\sum k^{p-2}c_k^{*p}} \le C\|x\|_{L_p}$$

c_k^* — the monotone rearrangement of the $|\ |$ coefficients.

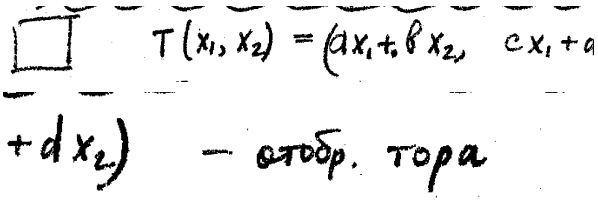
For monotone decreasing Fourier coefficients — two-sided inequalities (Hardy–Littlewood)

Between L_1 and $L_{1,5}$ there is no Hilbert interpolation space (Semyonov)

Gelfand's question: an abstract description of interpolation spaces. (it is unknown whether they are spaces of functions.)

Ya. Sinai.

Construction of invar. measures of smooth maps Markov partitions



Original source details: GELFAND-P000221

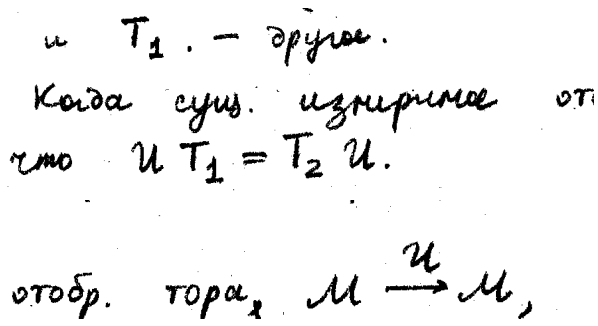
Diagram and typed description.



$T(x_1, x_2) = (ax_1 + bx_2,\ cx_1 + dx_2)$ — a map of the torus

and T_1. — another.

When does there exist a measurable map of the torus, $M \xrightarrow{U} M$, such that $UT_1 = T_2U$.



Original source details: GELFAND-P000221

Uncertain reading. First T, then T_1; in the equation, T_1 and T_2. The sign on the arrow and on both sides of the equation is looped: U / u; candidate: U, with letter case undetermined.

Irrational eigenvalues $\Rightarrow$ real $\Rightarrow$ ergodic, the spectrum is Lebesgue of countable multiplicity.

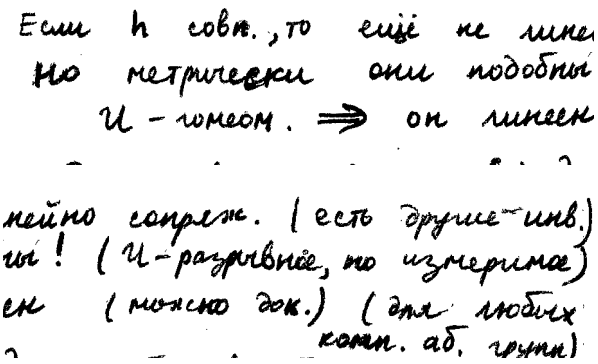
Entropy $h(T) = \ln \lambda \quad (\lambda > 1)$

GELFAND-P000222

If h agrees, this does not yet give linear conjugacy (there are other-inv.)

But metrically they are similar! (U — discontinuous, but measurable)

U — a homeom. $\Longrightarrow$ it is linear (one can prove this) (for any compact abelian groups)



Original source details: GELFAND-P000222

Uncertain reading. The looped map sign in both lines: U / u. Candidate: U; letter case is undetermined.

Ω_Π — the space of seq. $\omega = \{i_n\}$ — sequences infinite in both directions.

$i_n \in \{1, \ldots, S\}$

$$\underset{\text{of order } S}{\Pi} = \|x_{ij}\|, \quad x_{ij} = 1, 0 \qquad \pi_{i_n i_{n+1}} = 1$$

(only such sequences ω are admissible.)

We shall assume that Π^k consists of positive entries.

Consider the shift (the shift transformation on Ω).

Example. $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ — does not satisfy the cond., but Ω — sequences only of 1 or

Uncertain reading. [Crossed-out sign: illegible.]



Original source details: GELFAND-P000222

only of 2

$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ — anything may be put in

$\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ — look into it.

on Ω_Π we would like to specify an invariant measure μ with max entropy:

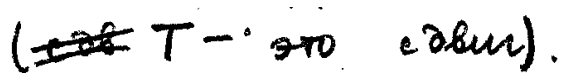


Uncertain reading. [An illegible crossed-out entry to the left of the formula.]

Original source details: GELFAND-P000222

$$h = \max_{\mu} h_\mu(T)$$

(



Uncertain reading. [Crossed out: where / shift.]

Original source details: GELFAND-P000222

T — this is the shift).

One must consider e with positive coordinates. $\lambda e = \Pi^* e$, $\lambda > 0$

(unique.)

(G. If all entries and all minors of order 2 are > 0, then there exist 2 eigenvectors, the 1st with positive entries, the 2nd — with the 1st change of sign —

— the matrix of minors of order 2 — a representation of the matrix itself in the exterior product.)

Consider a Markov chain with probabilities $p_{ij} = \dfrac{e_j \pi_{ij}}{\lambda e_i}$ —

[At the right:] — entropy — $\ln \lambda$

[In general, the entropy of a Markov chain

$$h = \sum_i e_i \left(- \sum_j \ln p_{ij}\, p_{ij} \right)$$

the measure of max entropy is unique.

The measure in Ω_Π on the Σ-alg. generated by cylindrical sets.

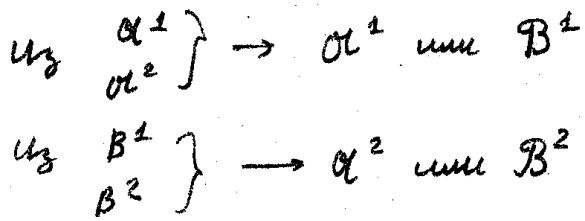
GELFAND-P000223

[source number: 10]

a) $\{a_1^1, \ldots, a_p^1;\ a_1^2, \ldots, a_q^2;\ b_1^1, \ldots, b_r^1;\ b_1^2, \ldots, b_s^2\}$
Π

$$\text{from } \left.\begin{matrix}\alpha^1\\ \alpha^2\end{matrix}\right\} \longrightarrow \alpha^1 \text{ or } B^1$$

$$\text{from } \left.\begin{matrix}\beta^1\\ \beta^2\end{matrix}\right\} \longrightarrow \alpha^2 \text{ or } B^2$$



Uncertain reading. In the transitions the shapes $\alpha/a/A$ and $\beta/b/B$ are not uniquely established; the B-like signs on the right are larger. The alphabet above has a and b. The displayed candidates are α, β on the left and B on the right.

Original source details: GELFAND-P000223

[In the right margin:] ‖ a partition is constructed from an integer matrix of order 2.

$$\lambda^2 - (p+s)\lambda + (ps - qr) = 0, \qquad \boxed{\boxed{ps - qr = 1}}$$

condition.

b) Ω_{Π_1} $\pi_{ij} = 1$ and $\pi_{N1} = 0$ (only in the 1st position)

Prop. The shifts in a) and b) are isomorphic.

[Everything now reduces to proving that
an algebraic automorphism ⇒
⇒ a partition.

Proof of the proposition.

Alphabet b): $A_1, \ldots, A_p$, $B_1, \ldots, B_s$.

(Isomorphism of any 2 Ω_Π with the same entropy).

Different groups are encoded (using the symbols A_i, B_i).

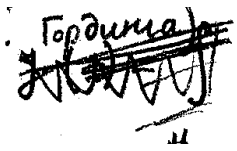
The space with the 1st max measure passes into a space with another max measure.

Next time: Markov partitions and how invariant measures are constructed with their help.

$23/\overline{\text{X}}$. Leray. Singularities of Feynman integrals.

N. Nilsson, Arkiv för Math. (Swede) 1964–1965

(student of Gording)



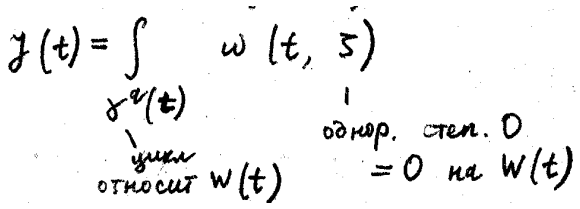
Undeciphered passage. A short formula below "(student of Gording)" is crossed out. Possible beginning: $J(\ldots)$; the remaining symbols are illegible.

Original source details: GELFAND-P000223



Original source details: GELFAND-P000223

Uncertain reading. A separate crossed-out sign below the formula: J / γ; its reading is undetermined.



Original source details: GELFAND-P000223

Uncertain reading. The cycle superscript is q / a; candidate: q. The coordinate sign in the argument of ω and below is ξ / ζ; candidate: ξ. The labels "cycle relative to $W(t)$", "homog. deg. 0", and "$= 0$ on $W(t)$" are readable.

$$J(t) = \int_{\gamma^q(t)} \omega(t, \xi)$$

cycle relative to $W(t)$ — homog. deg. 0 $= 0$ on $W(t)$

$t \in T = C^n$,

$\xi \in C^{\ell+1}$

$\omega(t, \xi)$ on P^ℓ, since it is homogeneous; the cycle is there too.

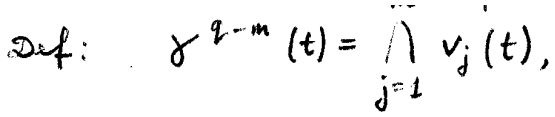
GELFAND-P000224

$W(t) = \bigcup_k W_k(t)$ — hyperplanes, algebraic functions of t, singularities may occur (not in general position)

$t \in S_0[W] \iff$ the dimension of the intersection of any number changes there.

Def:

$$\gamma^{q-m}(t) = \bigcap_{j=1}^{m} V_j(t),$$



Uncertain reading. Large sign: $\bigcap$ / $\bigwedge$; displayed candidate: $\bigcap$.

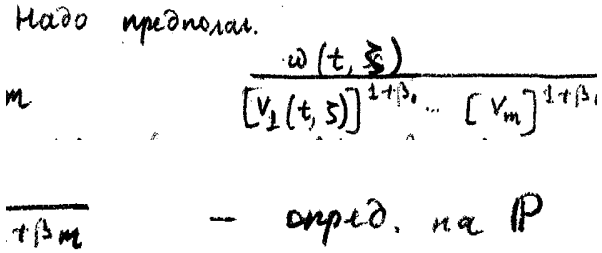
Original source details: GELFAND-P000224

$$V_j(t) = v_j(t, x) = 0$$

algebraic variety

One must assume

$$\frac{\omega(t, \xi)}{[v_1(t, \xi)]^{1+\beta_1} \dots [v_m]^{1+\beta_m}}$$ — defined on P

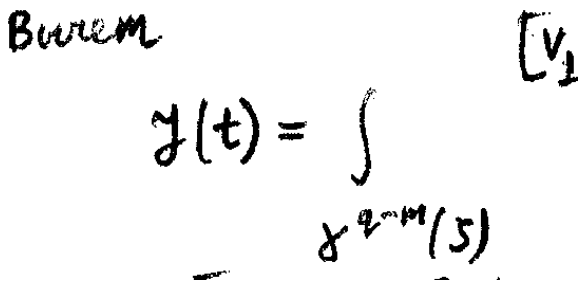


Uncertain reading. The coordinate signs in the argument of ω and first denominator: ξ / s; displayed candidate: ξ. The last factor is written $[v_m]^{1+\beta_m}$.

Original source details: GELFAND-P000224

[On the left:] Residue

$$J(t) = \int_{\gamma^{q-m}(\xi)} \frac{d^{\beta_1+\ldots+\beta_m}\omega}{[dv_1]^{1+\beta_1} \wedge \ldots \wedge [dv_m]^{1+\beta_m}}$$



Uncertain reading. Argument below the integral: ξ / s; displayed candidate: ξ.

Original source details: GELFAND-P000224

The singular support is also described in the general case

(On the covering space of the complement the form will be holomorphic)

Thm. Then $J(t)$ has algebraic singular support.



Uncertain reading. Right-margin note: “of type st. algebraic” / “of type sm. algebraic”; the first word and middle abbreviation are not uniquely established.

Original source details: GELFAND-P000224

(i.e. in T there exists an algebraic variety such that $J(t)$ is holomorphic on a covering of the complement.)

Leray described the singular support in the case of planes.

30/X̄ Sinai.

U-diffeom.

characteristic exp growth of the distance between nearby points (separation)

either as $t \longrightarrow +\infty$, or as $t \longrightarrow -\infty$.

$$\frac{d(T^n x, T^n y)}{d(x,y)} \geq a e^{cn} \qquad \begin{matrix} n > 0 \\ \text{or } n < 0 \end{matrix} \qquad (x, y \text{ — nearby})$$

Consider the points $d(T^n y, T^n x) \leq a e^{-cn} d(y, x)$,

Suppose that $\forall\, x$ this is a k-dimensional submanifold.

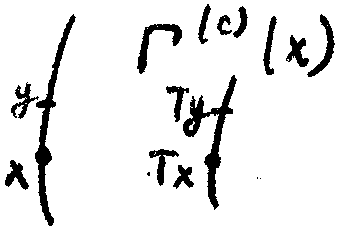
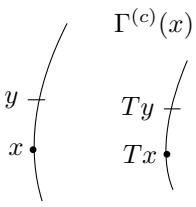


Diagram and typed description.



Original source details: GELFAND-P000224

This is a contracting submanifold (hence the index $\underline{c}$)

$$d(T^n y', T^n x) \leq a e^{+cn} d(y', x) \qquad (n \longrightarrow -\infty).$$

Similarly we obtain $\Gamma^{(p)}(x)$ — an expanding submanifold.
(ℓ-dimensional).
Then the choice does not depend on a and c.

GELFAND-P000225

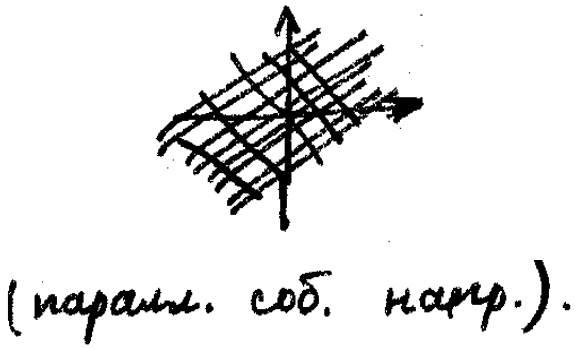
[source number: 11]

U-system: $k+\ell=n$ and the tangent spaces to the submanifolds are uniformly transverse

1. $k+\ell=n$
2. transversality.
3. Smoothness (not written out explicitly).

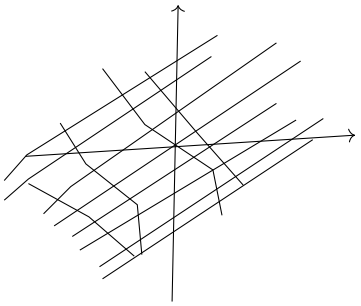
We obtain 2 foliations $\gamma^{(c)}$ and $\gamma^{(p)}$.
Example: linear transformation of the torus $(x,y)\longrightarrow(ax+by, cx+dy)$.

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = 1.$$



Original source details: GELFAND-P000225

Uncertain reading. Two families of inclined lines and axes with arrows upwards and rightwards. The number of coinciding lines and continuations within dense crossings are not uniquely established. Individual lines have no labels.

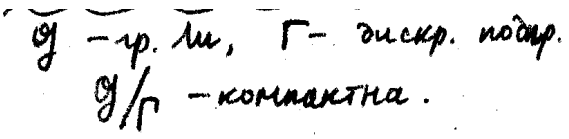


(parallel to the eigendirections).

A nonlinear perturbation (periodic and smooth) is also a U-diffeomorphism and the rotation number is the same.

Method of constructing U-diffeomorphisms.

g — a Lie group, Γ — a discrete subgroup.
g/Γ — compact.



Original source details: GELFAND-P000225

Uncertain reading. Lie-group notation: g / $\mathfrak{G}$; displayed candidate: g, distinct from the subsequent G.

T_{alg} — an algebraic automorphism of g, leaving Γ fixed $\quad(T_{\text{alg}}\Gamma=\Gamma)$

automorphism as a group

Then it acts on g/Γ.
Consider G — the Lie algebra; $\hat{T}_{\text{alg}}$ — the corresponding automorphism of G; if its
[At the left edge of the following line:] eigenvalues.
Among them there are no



Original source details: GELFAND-P000225

Uncertain reading. Crossed out: $\Longleftrightarrow$ / $\Longrightarrow$; endpoint directions are not established.

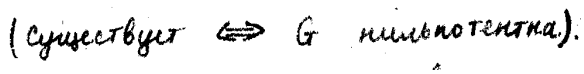
eigenvalues $= 1$ in mod



Original source details: GELFAND-P000225

Uncertain reading. Sign after mod: $\Leftrightarrow$ / a crossed-out sign.

$\Longleftrightarrow$ this is a U-diffeomorphism.
(They exist $\Longleftrightarrow$ G is nilpotent).



Original source details: GELFAND-P000225

Uncertain reading. Initial word in the parenthesis: “They exist” / “It exists”; the first candidate is displayed. The original fragment contains the entire parenthesis.

Further. A small nonlinear perturbation $T = T_{\text{alg}} + \Delta T$ is also a U-diffeomorphism.
Main problem: topological similarity to the algebraic one.
For small perturbations — from Anosov's general theorem
(A small perturbation of a U-diffeomorphism is topologically similar to the original).
Always.

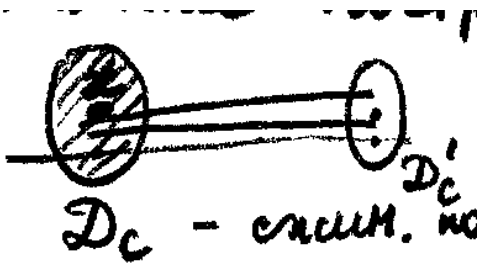
$$M = M_1 \cup M_2$$

M_2 — wandering points
M_1 — nonwandering points. — the condition U holds on them
[In the right margin:]

| also a
| conjecture

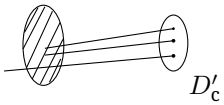
GELFAND-P000226

An important construction:



Original source details: GELFAND-P000226

Diagram and typed description.

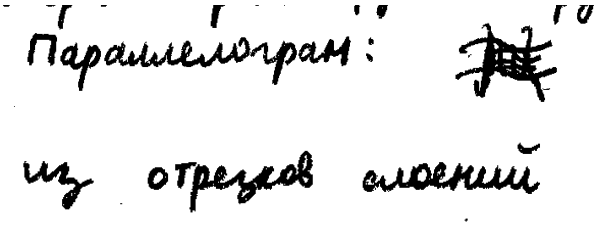


natural isomorphism of our pieces. of contracting leaves (along noncontracting leaves).

D_{c} — a contracting submanifold, dim $= k$.

(π — the canonical isomorphism.)

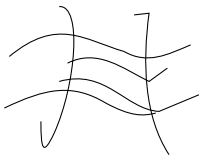
π is not always smooth (!), but it is absolutely continuous, i.e. the natural measures pass into one another.



Original source details: GELFAND-P000226

Uncertain reading. Two nearly vertical curved strokes and transverse curves; the number of coinciding strokes and their continuations at the dense middle crossings are not uniquely established. No labels or arrows are visible.

A parallelogram:



made of segments of foliations

(all its leaves are canonically isomorphic: both contracting and expanding.)

$$\Gamma(U) = \Gamma_{\mathrm{p}}(U) \cup \Gamma_{\mathrm{c}}(U)$$

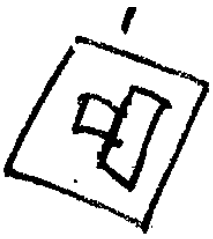
Suppose there is a system of parallelograms U_i adjoining one another.

$$M = \bigcup \overline{U}_i$$
$$U_i \cap U_j \subset \Gamma(U_i) \cap \Gamma(U_j) \qquad (\Gamma \text{ — boundaries}).$$

α — such a partition into parallelograms. Consider the boundaries:

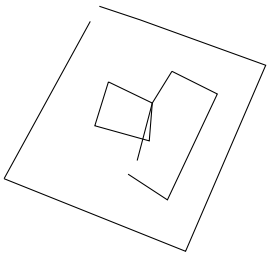
$$\Gamma_{\mathrm{p}}(\alpha) = \bigcup_i \Gamma_{\mathrm{p}}(U_i)$$
$$\Gamma_{\mathrm{c}}(\alpha) = \bigcup_i \Gamma_{\mathrm{c}}(U_i)$$

[Drawing on the right:]



Original source details: GELFAND-P000226

Diagram and typed description.

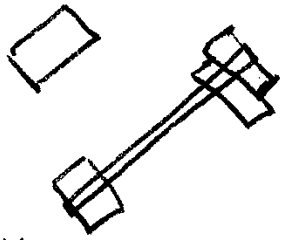


This α is called Markov if there exists an m such that

$$\boxed{\begin{array}{l}\Gamma_{\mathrm{p}}(T^{-m}\alpha) \subset \Gamma_{\mathrm{p}}(\alpha) \\ \Gamma_{\mathrm{c}}(T^{+m}\alpha) \subset \Gamma_{\mathrm{c}}(\alpha)\end{array}}$$

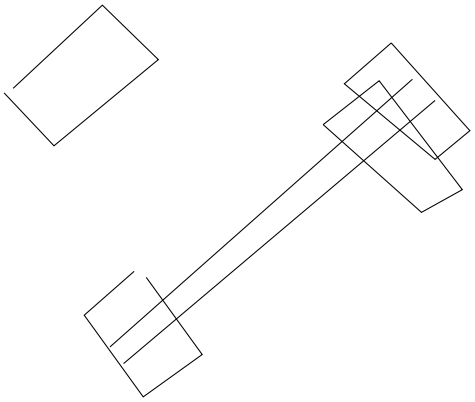
Thm. (technical) Markov partitions exist and there are many of them: $\forall$ α / partition into parallelograms , $\forall \varepsilon > 0$ $\exists m$ and a partition $\widetilde{\alpha}$, such that it will be at a distance of less than ε from the original.

[On the left:]



Original source details: GELFAND-P000226

Uncertain reading. A separate inclined quadrilateral and a narrow inclined strip with transverse quadrilaterals at its ends. The continuations of several superposed contours at the upper right end are not uniquely established. There are no arrows.



the ends cannot be in the middle.
(original parallelogram.)

— • —

If α is a partition with m, then one can make $m = 1$, taking the intersection

$$\alpha_1 = \alpha \cdot T^{-1}\alpha \cdot \ldots \cdot T^{-m+1}\alpha \text{ —}$$

— Markov with $m = 1$

GELFAND-P000227

[source number: 12]

— from

$$\Gamma_{\mathrm{p}}(\alpha_1) = \Gamma_{\mathrm{p}}(\alpha) \cup \Gamma_{\mathrm{p}}(T^{-1}\alpha) \cup \cdots \cup \Gamma_{\mathrm{p}}(T^{-m+1}\alpha)$$

Symbolic dynamics for a U-diffeomorphism.

(Suppose there is a Markov partition and with $m = 1$.)

$$\pi_{j_1 j_2} = \begin{cases} 1 & V_{j_1} \cap TV_{j_2} \neq \varnothing \\ 0 & \text{in the remaining cases} \end{cases}$$

Π

Ω_Π — the space of points

$$\omega = \{\dots\ j_{-n}, \dots, j_0, \dots, j_n, \dots\}$$
$$\pi_{j_n j_{n+1}} = 1.$$

Now associate

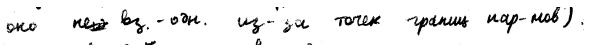
$$\omega \longrightarrow \bigcap_{-\infty}^{+\infty} T^n \overline{V}_{j_n} = x \in M \qquad (!)$$

(nonempty by the Markov property and, for a fine partition, consists of 1 point.)

We obtain a map $\varphi : \Omega_\Pi \longrightarrow M$

[A parenthesis begun on the right:] (from nonuniqueness.

it is not



Original source details: GELFAND-P000227

Uncertain reading. After the readable “not” a short entry is crossed out; possible “one-to-” is not established.

one-to-one because of the points — boundaries of parallelograms).

On a set of the 2nd category it is one-to-one.

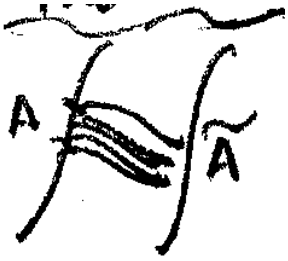
Condition: Π^K consists of positive entries for some K.

(i.e. all leaves $\Gamma^{(c)}$ and $\Gamma^{(p)}$ are dense in M — this is necessary and sufficient.)

On each leaf $\Gamma^{(c)}$ a σ-finite measure λ_c is introduced,

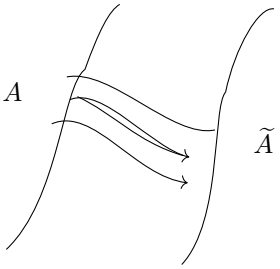
where

[On the left:]



Original source details: GELFAND-P000227

Uncertain reading. The lines labelled $A, \widetilde{A}$ and connecting curves. Directed thickenings are visible at two lower right ends; the number of coinciding lower curves and character of the other ends are not uniquely established.



1) $\lambda_c(A) = \lambda_c(\widetilde{A})$

(measures of canonically isomorphic sets are equal.)

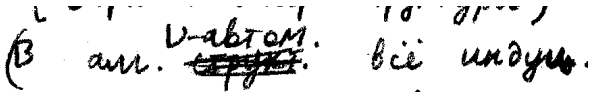
2) $\lambda_c(TA) = h\lambda_c(A) \quad (h < 1)$

(h — an algebraic number.)

— additional information.

(Reflects the homogeneous structure.)

(In an algebraic



Original source details: GELFAND-P000227

Uncertain reading. A word is crossed out, possibly “case”; its reading is not established.

[Inserted above:] U-automorphism everything is induced by Haar measure.)

Corollary:

there is an invariant measure, T is



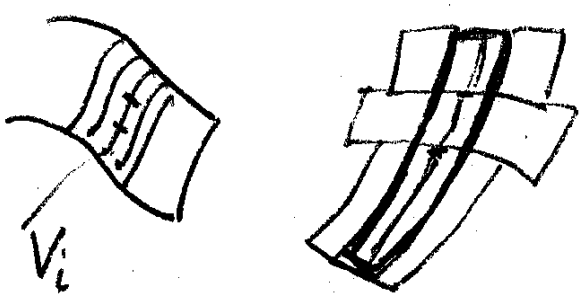
Original source details: GELFAND-P000227

Uncertain reading. Crossed-out sign: Cyrillic “б” / “в”; its reading is not established.

a finite Markov chain.

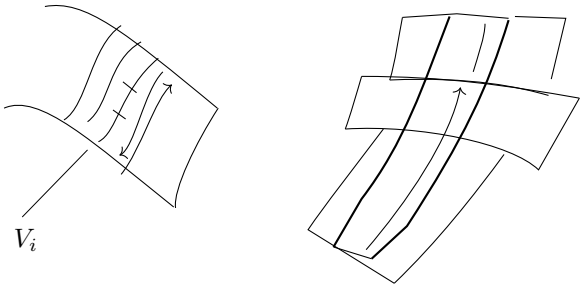
Construction of this measure:

$(n = 2)$

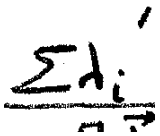


Original source details: GELFAND-P000227

Uncertain reading. Two diagrams with V_i, leader stroke, transverse marks, and visible directed strokes. Connections of thick lines at the base of the right diagram and some endpoint directions remain undetermined. There is no arrow between the diagrams.



$$\lambda_i h^{-1} = \sum \lambda'_i = \sum \pi_{ii'} \lambda_{i'}, \quad \text{i.e.}$$



Original source details: GELFAND-P000227

Uncertain reading. In the first sum the prime is high: $\sum \lambda'_i$ / $\sum \lambda_{i'}$; the first candidate is displayed.

$$\boxed{h^{-1}\vec{\lambda} = \Pi\vec{\lambda}.}$$

— eigenvector.

(This is for our



Uncertain reading. A short crossed-out entry is illegible.

Original source details: GELFAND-P000227

parallelograms).

The values of the measure on a segment are given. Then from the 2nd property.

[Below the last line:] (on a smaller segment.)

GELFAND-P000228



Uncertain reading. Энтропия / Энтропии (entropy); the ending is not uniquely established.

Original source details: GELFAND-P000228

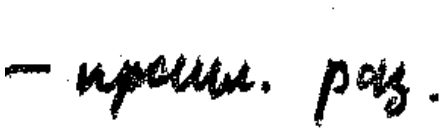
$$T = \begin{vmatrix} a & \cdot & b \\ c & & d \end{vmatrix}$$

$$h(T) = \ln \lambda \quad (\lambda > 1)$$

$$h(T) = \lambda$$

$\Rightarrow$ the transformations are metrically isomorphic.

Ω_{Π_1} maps onto Ω_{Π_0} (stand.) —

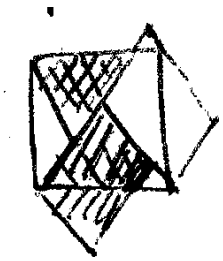


Uncertain reading. прежн. раз. / прим. раз. (prev. time / example, time); the first abbreviation is undetermined.

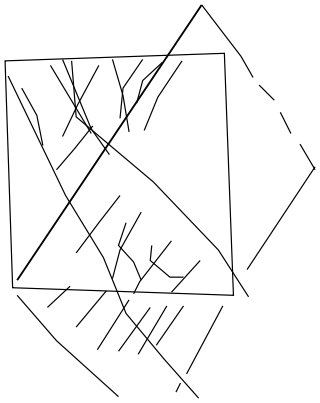
Original source details: GELFAND-P000228

And it remains to map the torus onto Ω_{Π_1}.

Π_1 is of a certain type: 4 groups of letters and properties of transitions — see earlier.

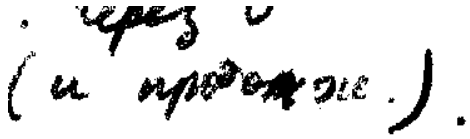


Uncertain reading. A nearly rectangular frame, oblique edges projecting above, to the right and below, and hatching. Continuations of lines at dense crossings and exact boundaries of the hatched parts are not uniquely established. There are no vertex labels or arrows.



Original source details: GELFAND-P000228

— a partition of the torus into 2 parallelograms (the sides ∥ the eigendirections)
The sides lie on a line through 0 in an eigendirection (and

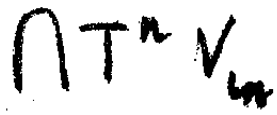


Uncertain reading. продолж. / проек. (cont. / proj.); the middle strokes of the abbreviation merge.

Original source details: GELFAND-P000228

).
This is a Markov partition (the coarsest). It is because of it that one succeeds in solving the isomorphism problem.

$$\bigcap T^n V_{\omega_n}$$

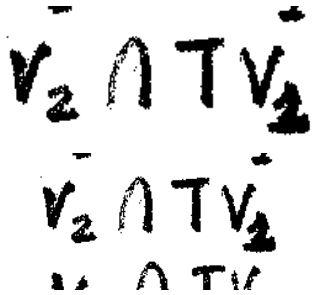


Uncertain reading. The subscript under V is ω_n / ω; a separate lower n is not established. No separate index is visible at the intersection sign.

Original source details: GELFAND-P000228

may consist of a large number of points.
Consider first 4 parallelograms:

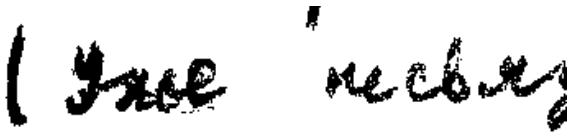
$$V_1 \cap TV_1$$
$$V_2 \cap TV_2$$
$$V_1 \cap TV_2$$
$$V_1 \cap TV_2$$



Uncertain reading. The second intersection: $V_2 \cap TV_2$ / $V_2 \cap TV_1$. The right subscript is heavily retraced; the first candidate is displayed.

Original source details: GELFAND-P000228

(



Uncertain reading. Уже (Already); the word is readable, but whether a transverse stroke crosses it out is undetermined.

Original source details: GELFAND-P000228

a disconnected partition. Form a partition — — its connected components. Their numbers, respectively — —



Uncertain reading. p, q, r, s; the second sign is heavily retraced and q remains a candidate.

Original source details: GELFAND-P000228

)

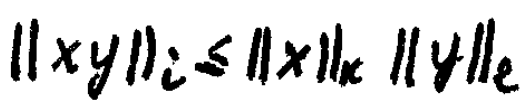
GELFAND-P000229

Year 1968.

12/II Question (Gelfand)

R — a countably normed ring:

$$\|xy\|_i \leq \|x\|_k \|y\|_\ell$$

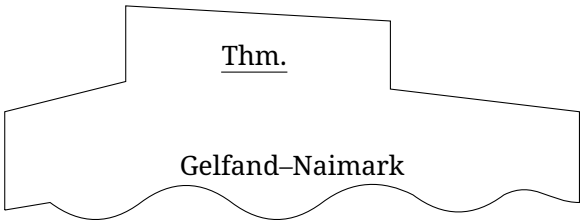


Uncertain reading. The last subscript: ℓ / e; ℓ is displayed.

Original source details: GELFAND-P000229



Diagram and typed description.



Original source details: GELFAND-P000229

for



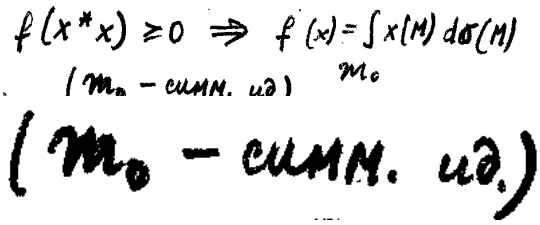
Original source details: GELFAND-P000229

Uncertain reading. [Superscribed sign: superposed loops and strokes; its reading is undetermined.]

normed rings:

$$f(x^*x) \geq 0 \Rightarrow f(x) = \int_{\mathcal{M}_0} x(M)\, d\sigma(M)$$

($\mathcal{M}_0$ — symm. ideals)



Original source details: GELFAND-P000229

Uncertain reading. In the lower limit and the explanation, the letter is M_0 / m_0 / $\mathcal{M}_0$; the lower zero is readable. Case and style are undetermined; $\mathcal{M}_0$ is displayed.

Extension — ? (Not always: one must characterize the class axiomatically).

$$x^*(M) = \overline{x(M')}, \quad M = M'$$

— symm.

L^1 with convolution, $\overline{x(-t)} = x^*(t)$

$\Rightarrow$ Bochner's theorem

$$\int_{-\infty}^{+\infty} e^{it\lambda}\, d\sigma(t)$$

I. Bernstein.



Original source details: GELFAND-P000229

Diagram and typed description.

Grothendieck topology.

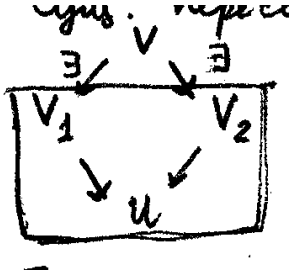
T is the following collection:

1) The category Cat T is specified (objects — open sets).

2) Certain families $\{\alpha_i : U_i \to U\}$ are specified, called coverings

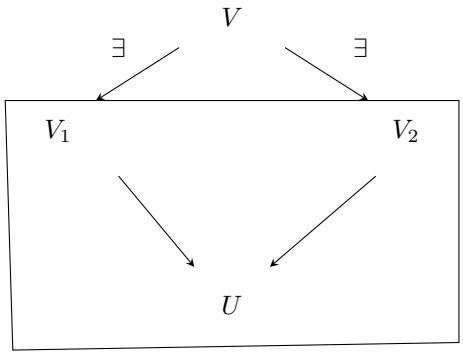
3) A system satisfying axioms is specified:

1. There exists an intersection — a product (fibered).

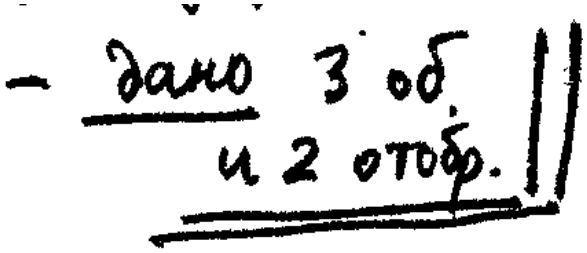


Original source details: GELFAND-P000229

Diagram and typed description.



(commut.).



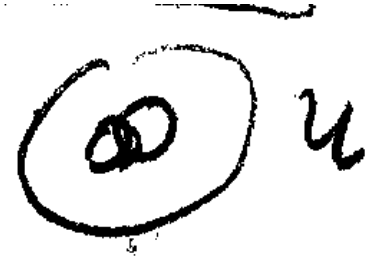
Original source details: GELFAND-P000229

Diagram and typed description.

— given 3 objects

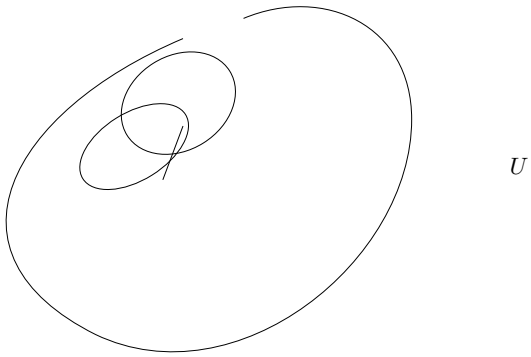
and 2 maps

$\boxed{V = V_1 \times_U V_2}$ — notation. Then there exists



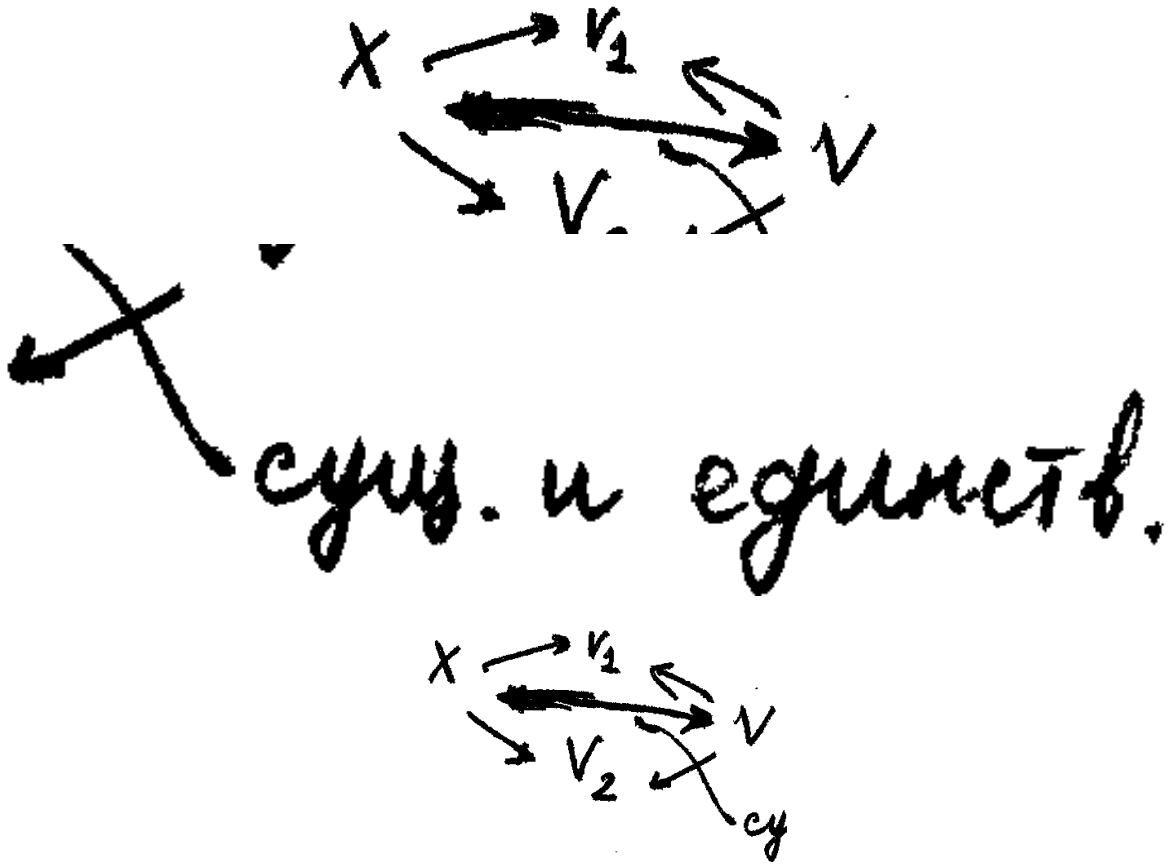
Original source details: GELFAND-P000229

Diagram and typed description.



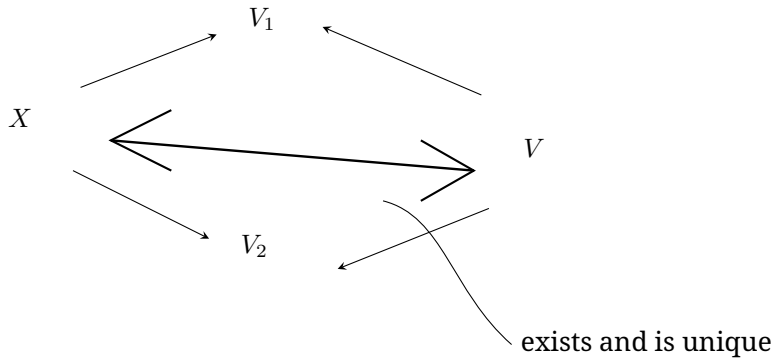
The universal property is required to hold:

$\forall X$, and $\forall$



Original source details: GELFAND-P000229

Uncertain reading. Four outer arrows: $X \to V_1$, $X \to V_2$, $V \to V_1$, $V \to V_2$. Wedge-shaped strokes are visible at both ends of the thick central connection; its direction and whether it was corrected are undetermined. The visible heads are retained but do not establish an unambiguous universal map. A leader points to “exists and is unique”.



Depends essentially on U (in Grothendieck top.)

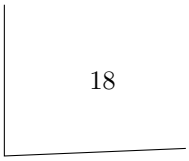
(If V exists, it is determined uniquely by the properties of the category)

GELFAND-P000230



Original source details: GELFAND-P000230

Uncertain reading. At the upper right, 18 / 98; the first digit is thickened. The candidate 18 and angular frame are displayed.



2. $\{U \xrightarrow{\text{id}} U\}$ — a covering.

3. $\{V_i \to V\}$ — a covering.

$\{V_{ij} \to V_i\}$ — a covering.

$\Rightarrow \{V_{ij} \to V\}$ — a covering.

4. $\{U_i \to U\}$ — a covering

$\{V \to U\}$ — a morphism

Then $\{U_i \times_U V \to V\}$

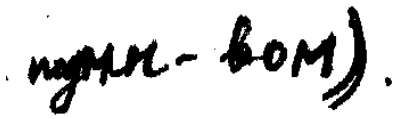
Example: unramified topology.

X — an ordinary topological space.

An open set: $U \to X$ — a local homeomorphism

(U — a topological space)

(a covering over X or its open

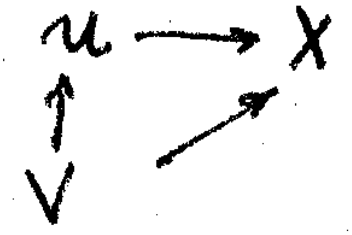


Original source details: GELFAND-P000230

Uncertain reading. подмн-вом (subset) / мн-вом (set), with possible cancellation of the initial “под”; no separate crossing-out stroke is established.

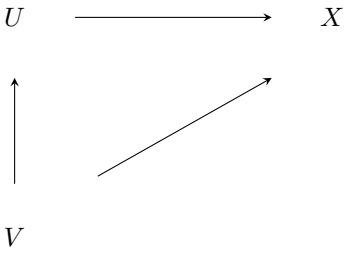
).

A map



Original source details: GELFAND-P000230

Diagram and typed description.



(any continuous one with a commutative diagram)

$\{U_i \to U\}$ — a covering, if the images



Original source details: GELFAND-P000230

Undeciphered passage. [The crossed-out word is undeciphered.]

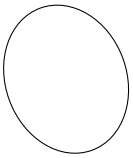
cover.

Axiom 1.



Original source details: GELFAND-P000230

Diagram and typed description.



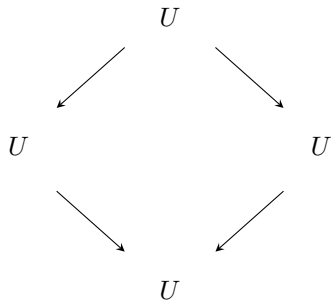
U — a circle, U — also a circle, a map of degree 2 $U \to U$.

$$U \times_U U = U.$$

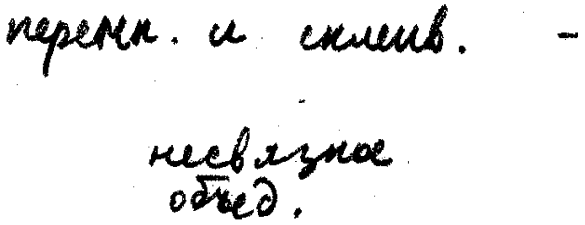


Original source details: GELFAND-P000230

Diagram and typed description.



$U_1 \times_X U_2$ —



Uncertain reading. перемн. и склеив. / пересеч. и склеив. (multipl. and gluing / intersect. and gluing); the first abbreviation is not uniquely established.

Original source details: GELFAND-P000230

— a disconnected union.

2 and 3 sheets $\Rightarrow$ 6 sheets

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \neq \mathbb{Z}_4,$$
$$\mathbb{Z}_2 \times \mathbb{Z}_3 = \mathbb{Z}_6.$$

GELFAND-P000231

Presheaf: • a functor

$$\text{Cat}\, T \to \text{Ab} \qquad \text{(contravar.)}$$
$$u \to P(u)$$

$$v \to u \;\Rightarrow\; P(u) \to P(v)$$

A sheaf — a presheaf satisfying the axioms:

1. $\xi \in P(u)$

$\{u_i \to u\}$ — a covering

$$\xi \to \xi_i \qquad P(u) \to P(u_i)$$

ξ is uniquely determined by ξ_i

2. and is indeed determined, if ξ_i agree on intersections

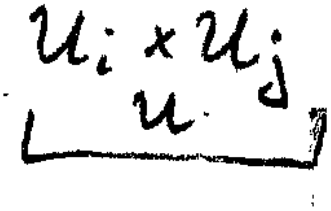


Diagram and typed description.

$u_i \times_u u_j$

Original source details: GELFAND-P000231

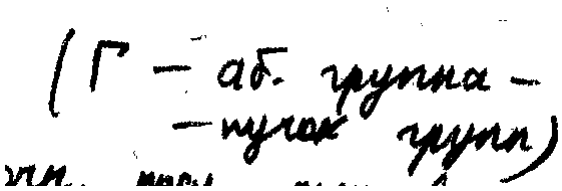
Define Čech cohomology

$$C^0 = P(u)$$
$$C^1 = \prod P(u_i)$$
$$C^2 = \prod P(u_i \times_u u_j).$$
$$C^3 = \prod P(u_i \times_u u_j \times_u u_k)$$

d is defined naturally

$\mathcal{F} \to \Gamma(X, \mathcal{F})$ — a functor.

(Γ — an abelian group —
— a sheaf of groups)



Uncertain reading. Both lines of the parenthesis are retained. No separate symbol is visible before the dash in the second line; its implied subject is undetermined.

Original source details: GELFAND-P000231

$$0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0.$$

— an exact sequence of sheaves.

$$0 \to \Gamma(\mathcal{F}_1) \to \Gamma(\mathcal{F}_2) \to \Gamma(\mathcal{F}_3)$$

(left exact).

Derived functors exist. $R^i : S \to \text{Ab}$

S — arbit. sheaves (category).

$$\to R^1(\mathcal{F}_1) \to \ldots \to R^2(\mathcal{F}_2)$$

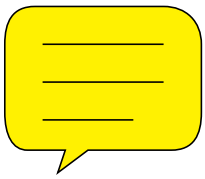
(R^i — determined uniquely).

$$R^i(\mathcal{F}) \underset{\text{def}}{=} H^i(X, \mathcal{F}).$$

Finite group. Grothendieck top. from representations.



Diagram and typed description. Separate icon on the source image.



Original source details: GELFAND-P000231

GELFAND-P000232

11/III Gelfand (and Kirillov)

$\mathfrak{g}$ — semisimple Lie algebra, (over C)

$U(\mathfrak{g})$ — its enveloping algebra

(polyn. in generators $e_1, \ldots, e_n$,

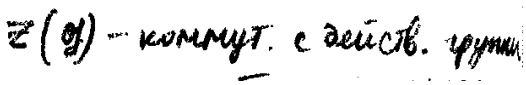
$$e_i e_k - e_k e_i = \sum C_{ik}^j e_j$$

$Z(\mathfrak{g})$ — centre of the enveloping algebra

(its elements — Laplace or Casimir operators).

Repr. of $G \Rightarrow$ diff. operators and commut.

$Z(\mathfrak{g})$ — commut. with the action of the



Uncertain reading. group

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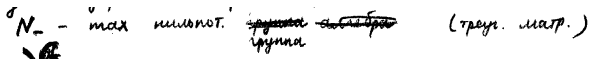
Such a repr. of the enveloping alg. by diff. operators is not necessarily faithful.

On G itself — faithful (transform.)

For $Z(\mathfrak{g})$ it makes no difference, left or right.

$\mathfrak{g}$ — semisimple Lie algebra. (over C).

N_- — max nilpot.



Uncertain reading. [crossed out: group algebra] [inserted below the first word: group]

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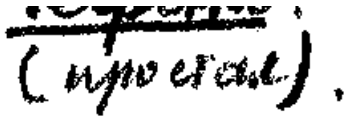
(triang. matr.)



Uncertain reading. $N_- \backslash G$ [upper right symbol heavily overwritten]

Original source details: GELFAND-P000232

Theorem: in $N_-\backslash G$ the repr. of $U(\mathfrak{g})$ is faithful.



Uncertain reading. (simple).

Original source details: GELFAND-P000232

f on the group is regular if its translates — finite-dim. space

(sin on the line)

On a comp. group, functions are approx. by regular ones (completeness theorem)

Consider functions $f(g) \equiv f(n_- g)$ — invar. — ring.

Consider Spec of this ring. This is X.

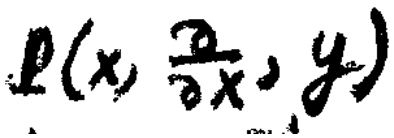
$N_-\backslash G$ birat. equiv. to projective space.

Enveloping algebras are numerous, but division rings, of fractions are few

$$\underset{\substack{\text{division ring}\\ \text{of quotients}}}{D(\mathfrak{g})} \sim D_{n,k}$$

Nontriv. that embeds into a division ring (!) of quotients.

$$x_1, \dots, x_n; \quad \frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n}, \quad y_1, \dots, y_k$$



Uncertain reading. $\ell(x, \frac{\partial}{\partial x}, y)$

Original source details: GELFAND-P000232

$\underline{\underline{QP^{-1}}}$ $Q_1 P_1^{-1}$ and so on. Everything can be done in 2 steps.

(QP^{-1} suffices)

n — number of variables, k — number of parameters — invariants.

GELFAND-P000233

[source number: 23]



Diagram and typed description.

23

Original source details: GELFAND-P000233

For a semisimple group it is almost proved that $D(\mathfrak{g}) \sim D_{n,k}$.

(



Original source details: GELFAND-P000233

Uncertain reading. постр.

is valid)

For nilpotent proved, and probably for algebraic solvable.

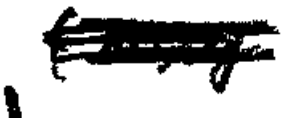


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Diagram and typed description.

Adjoin to $Z(\mathfrak{g})$ (K) operators — virtual Laplace operators

New ring $\tilde{U}(\mathfrak{g})$ — with adjoined left translations



Original source details: GELFAND-P000233

Undeciphered passage. A short word or abbreviation is crossed out; its letters beneath the heavy strokes have not been established.

$$U(\mathfrak{g}) \otimes_{Z(\mathfrak{g})} \tilde{Z}(\mathfrak{g}) = \tilde{U}(\mathfrak{g}).$$

$\tilde{Z}$ — Z with adjoined virt. Laplace operators.

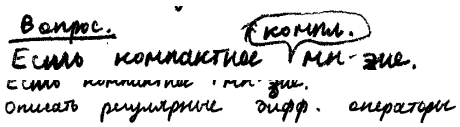
(this is an algebraic extens., Galois group — Weyl group).

Theorem. The division ring of fractions of $\tilde{U}(\mathfrak{g})$ is $D_{n,k}$.

(For U unknown; done for unimodular).

Alg. extens. of $D_{n,k}$ — ?.

In the commut. case, alg. varieties.



Original source details: GELFAND-P000233

Diagram and typed description.

Question. There are compact

compl. manif.

Describe regular differential operators

On the Riemann sphere $\frac{\partial}{\partial z}$, $z\frac{\partial}{\partial z}$, $z^2\frac{\partial}{\partial z}$

On Dixmier's work (Limanov).

$R_{1,0}$ — ring of quantum mech. — generators x, $\frac{d}{dx}$.

Diagram and typed description.

Polynomials

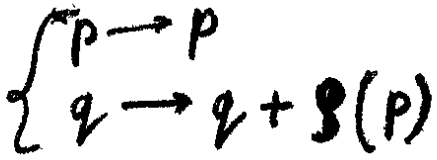
in p and q.



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$p, q, \quad [p, q] = 1.$

The autom. group is generated by triang. autom.



Uncertain reading. $\begin{cases} p \to p \\ q \to q + f(p) \end{cases}$

Original source details: GELFAND-P000233

and $\begin{cases} p \to p + g(q) \\ q \to q \end{cases}$

For the division ring $D_{1,0}$ unknown.

Or



Uncertain reading. [crossed-out symbol: и? от?]

Original source details: GELFAND-P000233

$\begin{cases} q \to q \\ p \to p + g(q) \end{cases}$

and $\begin{cases} q \to p \\ p \to -q \end{cases}$

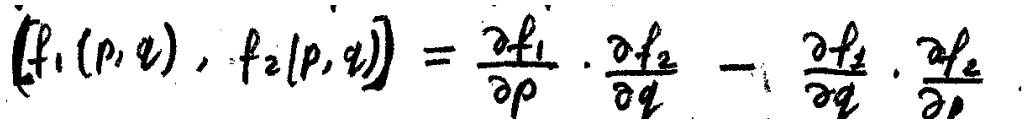
(Fourier transform)

Is generated

GELFAND-P000234

Arnold: theor.

Let there be functions $f(p, q)$



Original source details: GELFAND-P000234

Uncertain reading. $[f_1(p,q), f_2(p,q)] = \frac{\partial f_1}{\partial p} \cdot \frac{\partial f_2}{\partial q} - \frac{\partial f_1}{\partial q} \cdot \frac{\partial f_2}{\partial p}$. [Index in the first factor after the minus: 1 or 2.]

Consider 2 polynomials

$$P = P(p, q)$$
$$Q = Q(p, q).$$

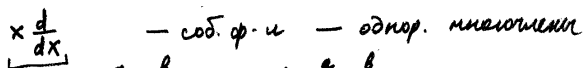
Substitute P and Q — will be 1.

They induce canon. transformations of quasiclassics.

But there is no connection, even with grading.

An element x is completely semisimple if the transformation ad X has a complete system of eigenvalues.

Then by an automorph. x is carried into $x \to pq$.



Original source details: GELFAND-P000234

Diagram and typed description.

$$\underbrace{x\frac{d}{dx}}_{\text{homog. polynomials.}} \quad \text{— eigenfunctions —}$$

$$p^a q^b \to (a-b)p^a q^b.$$

Result of



Original source details: GELFAND-P000234

Uncertain reading. Амицури or Амицура

:

x $C(x)$ — commuting — commut. subalgebra.
But these are not always polynomials.

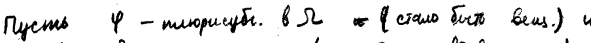
$$\forall \alpha \; \exists X, Y, \qquad X^3 = Y^2 + \alpha, \qquad \alpha \in C.$$

Palamodov.
on Hörmander's work (on $\bar\partial$)
2 main results: existence theorem and approx. theorem.
1. $\Omega \subset C^n$, $\phi \in C^2(\Omega)$

$$L(\phi) = \frac{\partial^2 \phi}{\partial z_j \partial \bar z_k} \xi_j \bar\xi_k \geq 0$$

$$L_{p,q} = L_{p,q}(\Omega, \phi), \qquad \|f\|^2_\phi = \int_\Omega |f|^2 e^{-\phi} dV$$

forms of type (p, q)
Let ϕ be plurisubh. in Ω



Original source details: GELFAND-P000234

Uncertain reading. [overlapping strokes before the parenthesis]

(therefore real) and
$\forall c \; \{\phi \leq c\}$ — compact. ($\Rightarrow \Omega$ — pseudoconvex)
Then $\forall f \in L_{p,q}$, $\bar\partial f = 0$
$\exists$



Original source details: GELFAND-P000234

Undeciphered passage. Crossed-out fragment; its initial symbol resembles g, the remaining letters have not been established.

g of type $p, q-1$

$$\bar\partial g = f \quad \overset{(q\geq 1)}{\text{and}} \quad g \in L_{p,q-1}(\Omega, \phi + 2\ln(1+|z|^2))$$

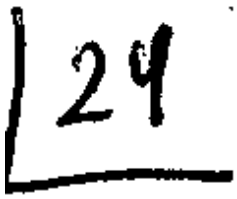
Here $\sqrt{q}\,\|g\| \leq \|f\|$ (assuming $q \geq 1$)

GELFAND-P000235

[source number: 24]

Diagram and typed description.

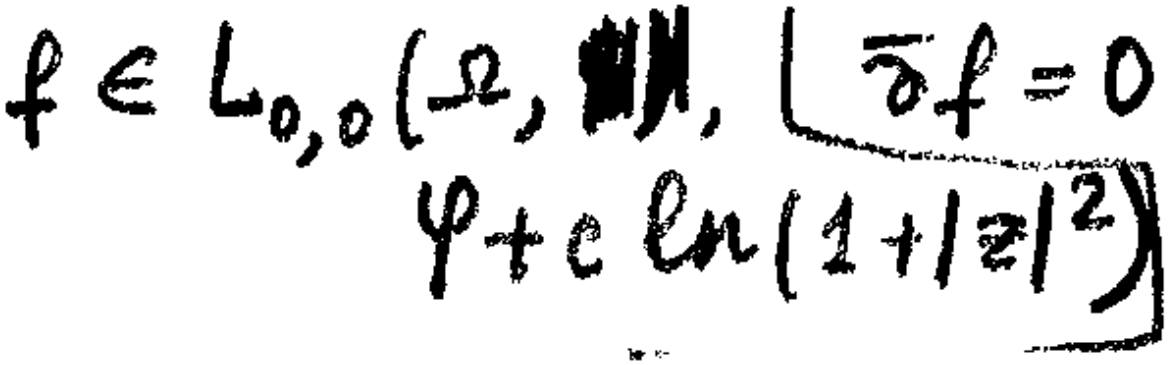
24



Original source details: GELFAND-P000235

2. Approx. theorem.

$\Omega = C^n$.



Original source details: GELFAND-P000235

Uncertain reading. The former second argument is crossed out: ϕ or ψ; $\phi + c\ln(1+|z|^2)$ is inserted below.

$f \in L_{0,0}(\Omega, [?])$ $\quad \bar{\partial} f = 0$ (holom. functions)

$\phi + c\ln(1+|z|^2)$

i.e. $|f| \le Ce^{\phi}(1+|z|^2)^k$, $(\exists C > 0)$.

This set is nonempty (nontriv. — $\Omega = C^n$). — and dense in the space of all [holom.] entire functions with unif. convergence on $\forall$ compact set:

ϕ — homog. along radii $\Rightarrow$ indicator of an entire function.

Hence Martino: the class of indicators — all radially homog. plurisubh. harm.



Original source details: GELFAND-P000235

Uncertain reading.

$$i(z) = \overline{\lim_{\tau\to\infty} \tau^{-k} \ln |f(z\tau)|}$$

The affiliation of the stroke above lim has not been established.

18/III

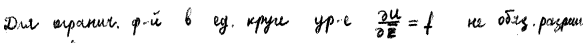
If $\phi = \phi(x)$ (only on the real parts), it is plurisubharm. if and only if it is convex.

In $\forall$ one-dim. complex space subharm.

Example:

$$\frac{\partial u}{\partial \bar{z}} = f, \qquad |f(z)| \le Ce^{a|\operatorname{Im} z|}(1+|z|^2)^k.$$

A solution in this class — nontrivial.
For bounded functions in the unit disk, the equation



Original source details: GELFAND-P000235

Uncertain reading. $\frac{\partial u}{\partial z} = f$ [denominator: z or $\bar{z}$]

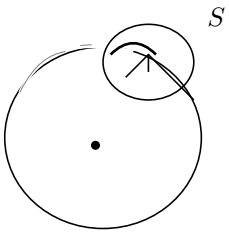
is not necessarily solvable for bounded f.

For solvability it suffices
(f — measure)



Original source details: GELFAND-P000235

Uncertain reading.



The faint upper arc and dark strokes at the intersection are transcribed without an established meaning for the strokes.

S — disk. $|f(S\cap\Omega)| \le CR_S$.

Theorem 3. More precise theorem for fixed q.

Let $\partial\Omega \in C^3$ and the domain be bounded. Local condition on the boundary:

$\forall z \in \partial\Omega\ \exists \phi \in C^2$ such that $\Omega = \{\phi(z) < \phi(z_0)\}$ in a nbhd. of this point.

Condition a_q:

$$\sum \frac{\partial^2 \phi}{\partial z_j \partial \bar{z}_k} \xi_j \bar{\xi}_k \quad \text{on} \quad \sum \frac{\partial \phi}{\partial z_j} \xi_j = 0$$

has —

$$\begin{cases} q+1 \text{ negative} \\ \text{or } n-q \text{ positive} \end{cases} \quad \| \quad \text{eigen-values}$$

Gelfand: domains of regularity — ?

Palam. says nobody is working on this.

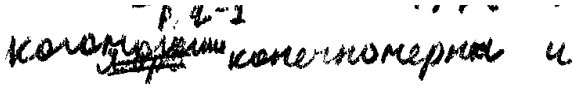
$\partial\Omega$ pseudoconvex if at $\forall$ point condition a_0 (at $\forall$ point $n-1$ positive eigenvalues)

In the tangent space, $n-q$-dim. surfaces along which the domain is pseudoconvex.

Let a_q hold at $\forall$ boundary point.

$$L_{p,q-1} \xrightarrow{\bar{\partial}} L_{p,q}(\Omega, \phi) \xrightarrow{\bar{\partial}} L_{p,q+1}$$

Cohomology



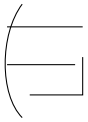
Uncertain reading. [crossed out below: kernel, ядро or ядра]

Original source details: GELFAND-P000235

is finite-dimensional and the image is closed.



Uncertain reading.

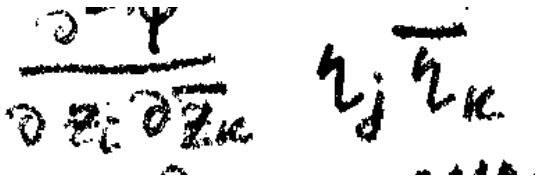


Isolated sign: E or $\exists$; its function has not been established.

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Existence of a function Ψ on the whole domain.

Ψ — real, defined inside Ω,



Uncertain reading. $\frac{\partial^2 \Psi}{\partial z_i \partial \bar{z}_k} \eta_j \bar{\eta}_k$ [first denominator index: i or j]

Original source details: GELFAND-P000235

satisfies all conditions

(1 more plus required, since in the whole space)

Then cohomology 0.

GELFAND-P000236

All these theorems for complex spaces — Andreotti and Grauert.

1. Runge-type theorem.

$\omega \subset \Omega$, $L(\Psi)$ at $\forall$ point of $\Omega \setminus \omega$ satisfies the condition

$$\begin{array}{c} q+1 \quad - \\ \text{or} \\ n-q+1 \quad + \end{array}$$

$\Psi \to \infty$ near $\partial\Omega$, $\{\Psi < 0\} = \omega$

Then the cohomology of Ω restricts monomorphically to the cohom. of ω.

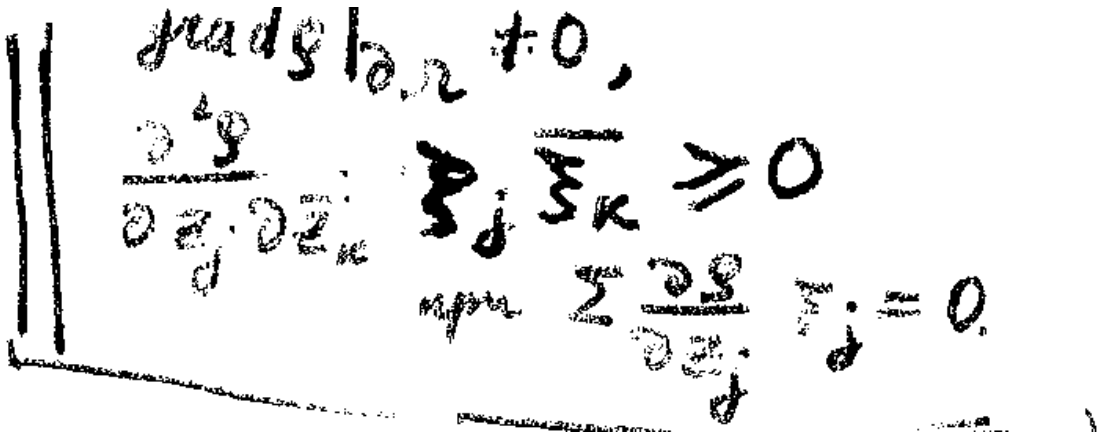
A closed form f in Ω and the equation $\bar{\partial} u = f$ solvable in $\omega \Rightarrow$ it is solvable in Ω as well.

2. Approximation theorem.

Closed forms in ω are approximated by closed forms in Ω.

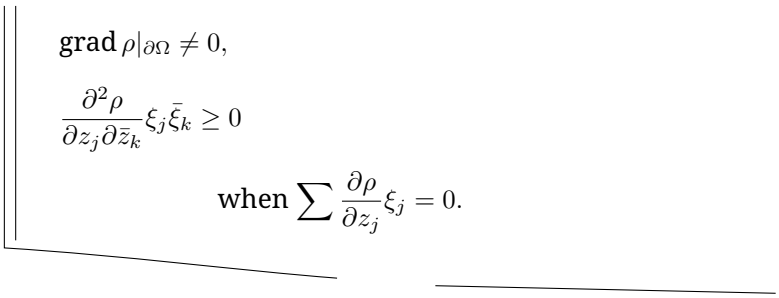
Existence theorem.

Let $\partial\Omega \in C^2$ and there exist ρ, $\{\rho < 0\} = \Omega$,



Original source details: GELFAND-P000236

Diagram and typed description.



It is proved simply, using the identity.

$$L_{p,q-1} \xrightarrow{\bar\partial=T} L_{p,q} \xrightarrow{\bar\partial=S} L_{p,q+1}$$

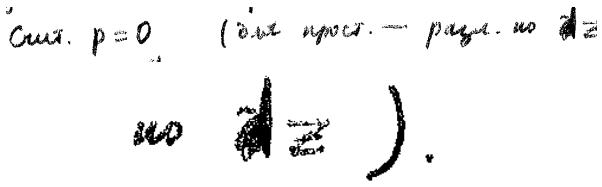
$$\|T^* f\|_\phi^2 + \|Sf\|_\phi^2 =$$

$$(u,v)_\phi = \int_\Omega u\bar v e^{-\phi} dV.$$

(adjoint with respect to this inner product)

$$= \sum_L \sum_{j,k} \left(\frac{\partial^2 \phi}{\partial z_j \partial \bar z_k} f_{jL}, f_{kL} \right)_\phi + \sum_J \sum_k \left\| \frac{\partial f_J}{\partial \bar z_k} \right\|_\phi^2 +$$

Take $p = 0$ (for simplicity — expansion in



Uncertain reading. dz [first symbol heavily overwritten; reading not established]

Original source details: GELFAND-P000236

).

$$f = \sum f_J d\bar z^J, \qquad J = j_1, \ldots, j_q.$$

f_J skew-symm. in the indices, jL — long index.

$$+ \sum \sum \int_{\partial\Omega} \frac{\partial^2 \rho}{\partial z_j \partial \bar z_k} f_{jL} \overline{f_{kL}} e^{-\phi} dS.$$

The identity — integration by parts.
First — Morri with $\phi = 0$.
grad, rot — known in mechanics.
Solvability follows from it.
Next time — Nirenberg (according to hearsay).